

Installation Training

AFL Wrapping Tube Cable (WTC)[®]

(Pulling / Air-Assisted Installations)



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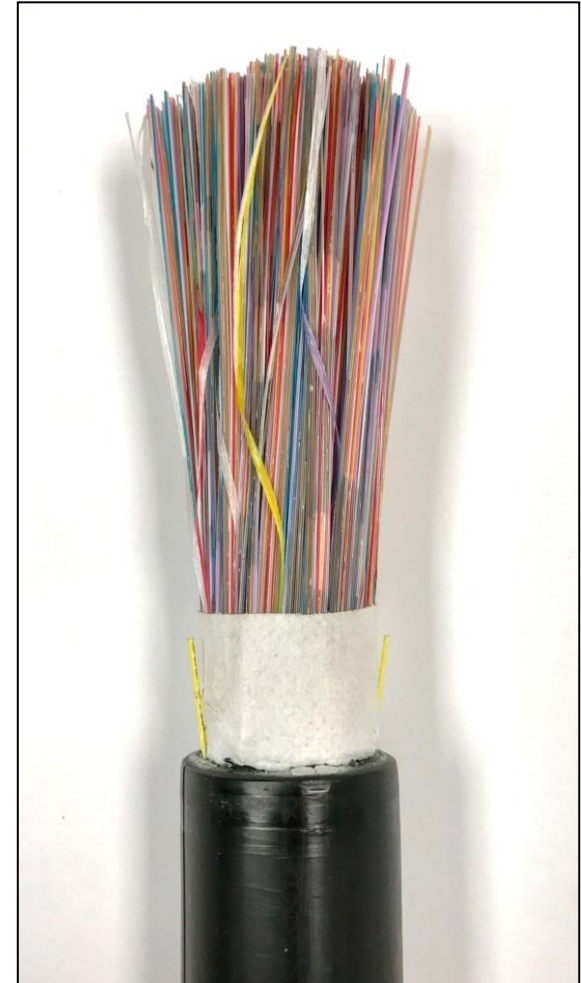
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Wrapping Tube Cable (WTC) Design and Specifications

WTC Design and Specification (Description)

General Description:

- Qualified to the GR-20 Specifications
- Pulled, Jetted (Blown), Lashed or Direct Buried
- AFL Offers Fiber Counts from 144F to 6912F
- Ultra High-Density Designs
- Small Diameter and Low Cable Weight
- Armored and Dielectric Versions
- Water Blocked Dry Core
- Fast Entry and Cable Prep
- Mass Fusion Splicing Efficiency
- Significant “Total Cost” Savings Impact



WTC Design and Specification (Cable)

AFL “Wrapping Tube Cable” (WTC) - Available in:

Non-armored 250 um: 144, 288, 432, 576, 864, 1152, 1728

Armored 250 um: 144, 288, 432, 576, 864, 1728

Non-armored 200 um: 1728, 3456, 6912

AFL SWR Fiber Binder Groups:

72 Fibers: (288 – 864) **144** Fibers: (1152 – 3456) **288** Fibers: (6912)

(Note: The **144**-fiber cable does not have binders (1 -12 Ring Marks)

The **288**-fiber **non-armored** does not have binders (1 – 24 Ring Marks)

Double Bound Binders:

Attached for Ease of Group Separation.

Water Blocking Tape: (Dry Core)

Water Protection and Bundling of the Fiber Groups.

Jacket Ripcords:

Installed on Both Single and Double Jacket Designs.

Strength Members:

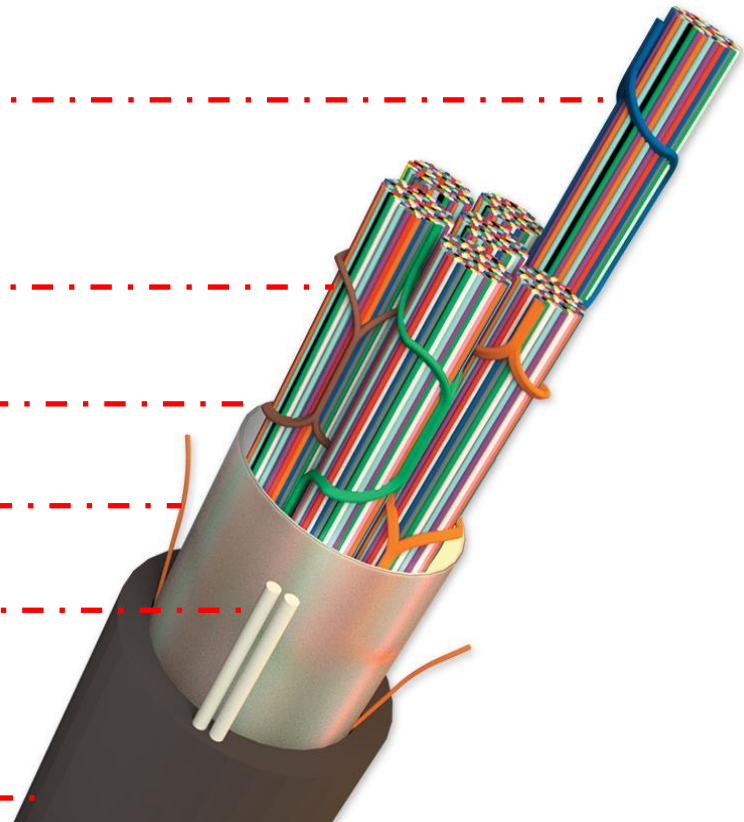
Two and Four Embedded Strength Members Designs used for Temperature Stability and Ease of Closure Installation.

144F: (1+1) and 288F– 6,912F (2+2)

Outer Jacket Sheath:

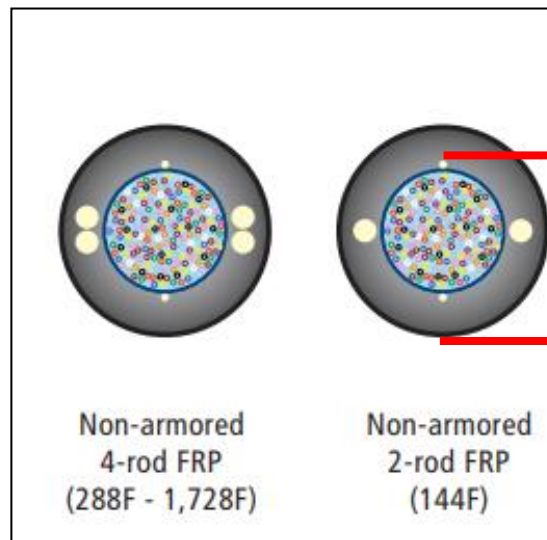
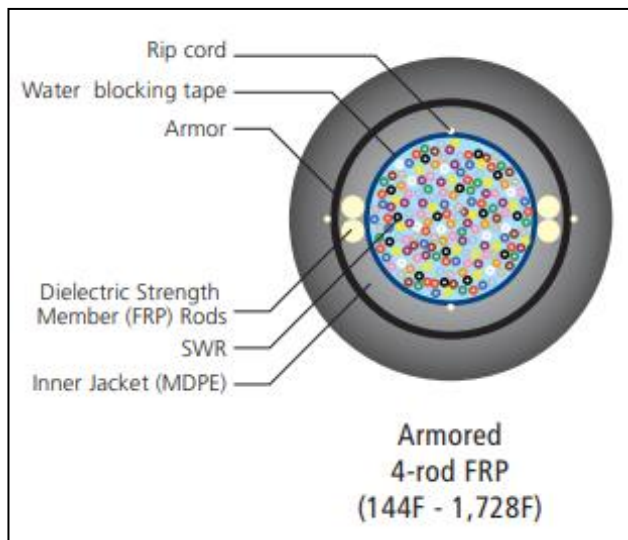
Both Single Jacket and Double Jacket Single Armor Designs.

Note: See design specifications for availability.

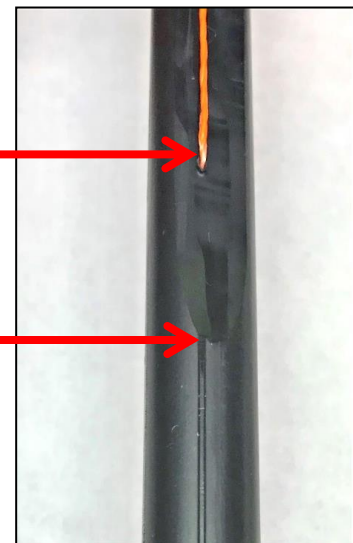


WTC Design and Specification (Cable)

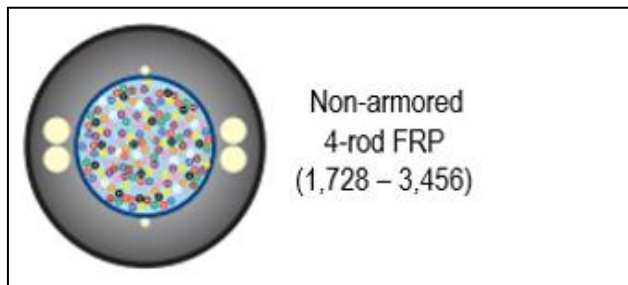
250 um Designs:



Ripcord - Ridge



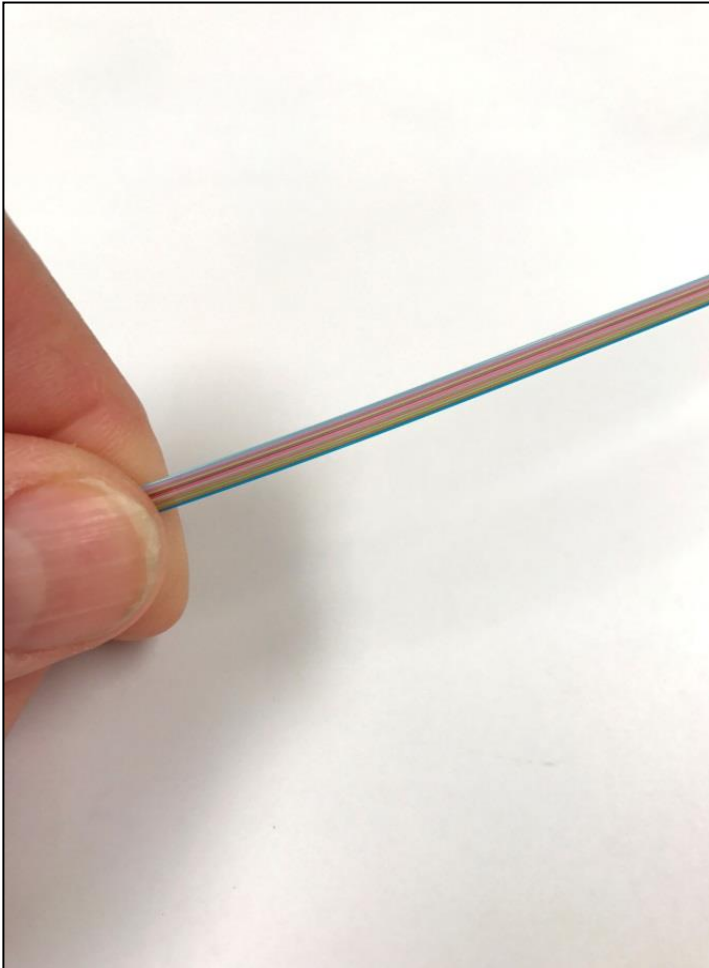
200 um Designs:



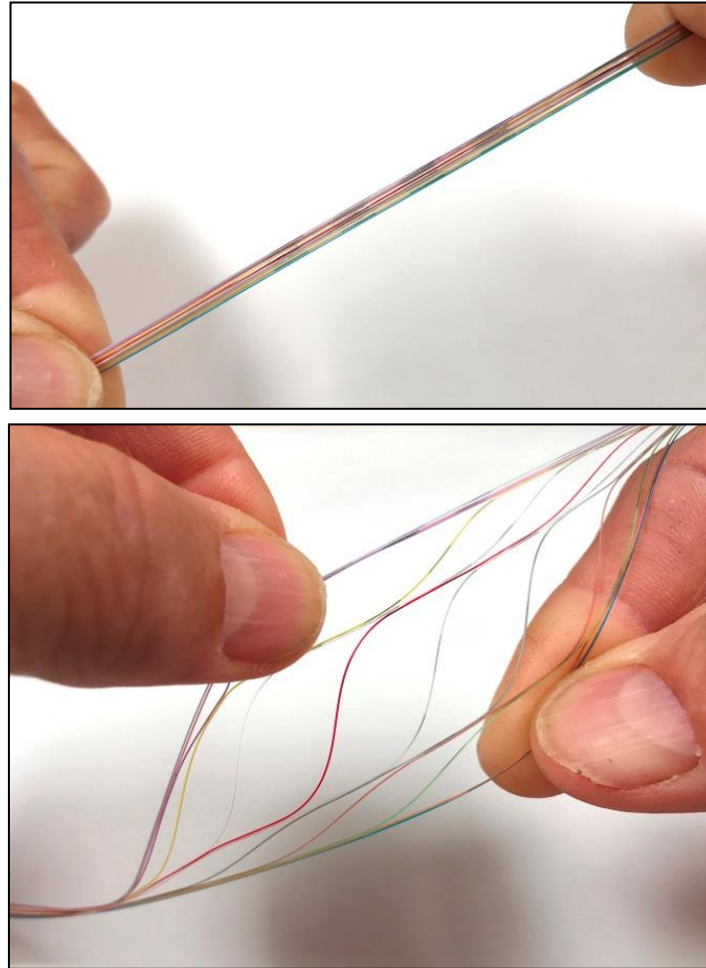
Note: A ripcord ridge is located on both the 250 um and 200 um Non-Armored WTC.

WTC Design and Specification (Ribbon Fiber)

Encapsulated (Matrix) Ribbon



AFL SpiderWeb Ribbon (SWR)



WTC Design and Specification (SWR Color Code)

SWR Color Code:

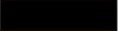
TIA-598 Rev. D Optical Fiber Cable Color Coding

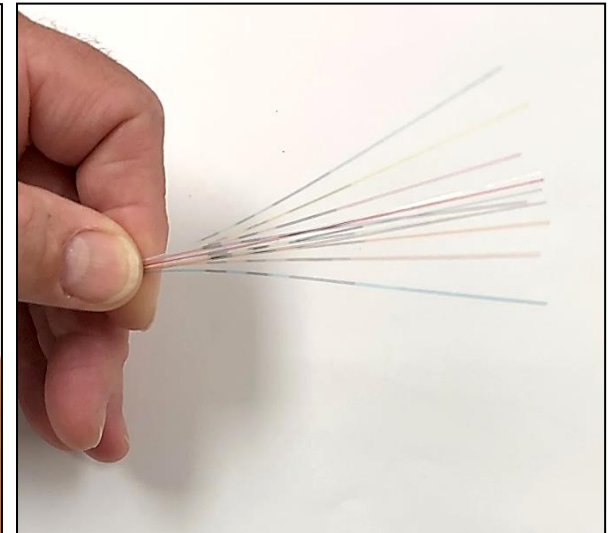
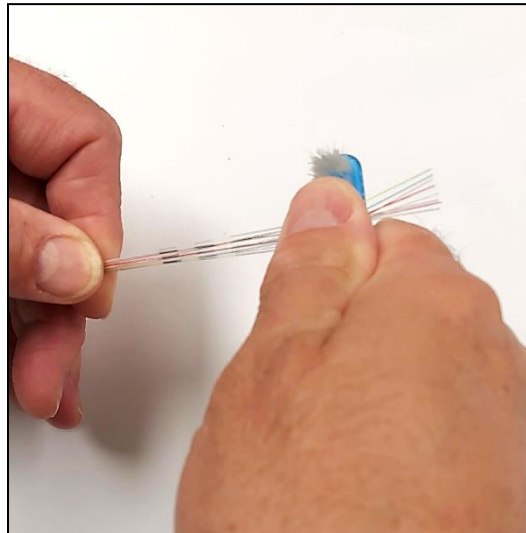
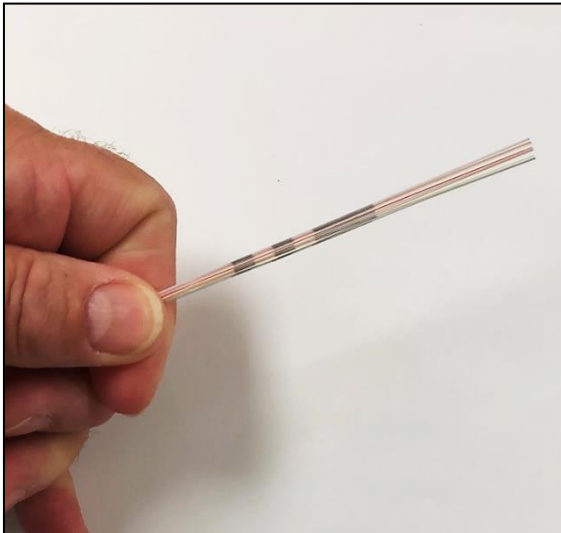
Each SpiderWeb Ribbon (SWR) contains **twelve (12)** **optical fibers** that are colored Blue thru Aqua.

Each fiber can be individually separated for **single splicing** with the use of a **soft bristled toothbrush**.

SpiderWeb Ribbon (SWR) Separation Procedure:

OPTICAL FIBER COLOR CODE

Fiber Number	Color	
1	Blue	
2	Orange	
3	Green	
4	Brown	
5	Slate	
6	White	
7	Red	
8	Black	
9	Yellow	
10	Violet	
11	Rose	
12	Aqua	



WTC Design and Specification (Fiber Type)

Optical Fiber Types (SWR) in WTC:

144 – 864: Fujikura **Ace** Fiber (**250 μm**)

1152 – 1728: Fujikura **SR-15e** Fiber (**250 μm**)

FIBER COUNT	FIBER DESIGNATOR	MFD	MAXIMUM ATTENUATION (CABLED) dB/km		
			1310 nm	1383 nm	1550 nm
144, 288, 432, 576, 864	9 (ITU-T G.652D/G.657.A1)	$9.2 \pm 0.4 \mu\text{m}$	≤ 0.40	≤ 0.40	≤ 0.30
1152, 1728	K (ITU-T G.652D/G.657.A1)	$8.6 \pm 0.4 \mu\text{m}$	≤ 0.40	≤ 0.40	≤ 0.30



1728 – 6912: Fujikura **SR-15e** Fiber (**200 μm**)

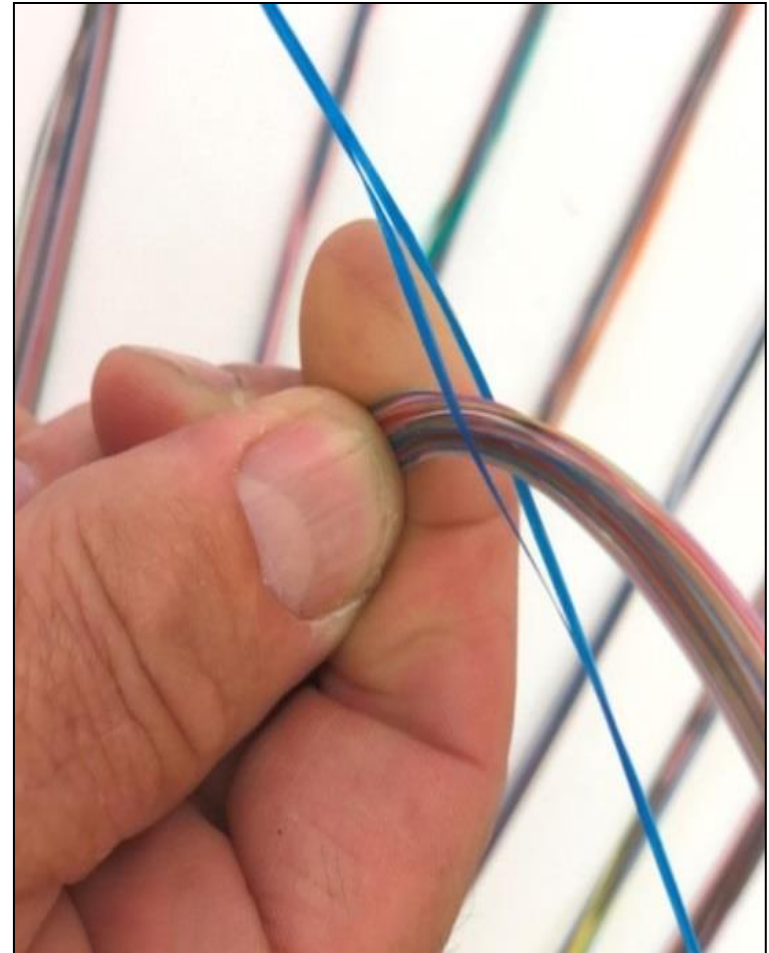
FIBER COUNT	FIBER DESIGNATOR	MFD	MAXIMUM ATTENUATION (CABLED) dB/km		
			1310 nm	1383 nm	1550 nm
1,728	BE (ITU-T G.652.D and G.657.A1)	$8.6 \pm 0.4 \mu\text{m}$	$\leq 0.4 \text{ dB/km}$	$\leq 0.4 \text{ dB/km}$	$\leq 0.3 \text{ dB/km}$
3,456	BE (ITU-T G.652.D and G.657.A1)	$8.6 \pm 0.4 \mu\text{m}$	$\leq 0.4 \text{ dB/km}$	$\leq 0.4 \text{ dB/km}$	$\leq 0.3 \text{ dB/km}$
6,912	BE (ITU-T G.652.D and G.657.A1)	$8.6 \pm 0.4 \mu\text{m}$	$\leq 0.4 \text{ dB/km}$	$\leq 0.4 \text{ dB/km}$	$\leq 0.3 \text{ dB/km}$

WTC Design and Specification (Fiber Groups)

Fiber Binder Groups



Dual Binder System



WTC Design and Specification (Fiber Group ID)

Fiber Group ID – 250 um:

1-12 Ring Marks: 144F

12 Ribbons / No Binder

288F Non-Armored: 1-24 Ring Marks / 24 Ribbons / No Binders

288F Armored: 1-6 Ring Marks / 72F SWR Binder Groups

432 – 864 Fiber: 1-6 Ring Marks

72F SWR Binder Groups

1152 – 1728 Fiber: 1-12 Ring Marks

144F SWR Binder Groups

R NO.	STRIPE RING MARKING	R NO.	STRIPE RING MARKING	R NO.	STRIPE RING MARKING	R NO.	STRIPE RING MARKING
1	█	7	███	13	██████	19	██████████
2	██	8	████	14	███████	20	███████
3	███	9	█████	15	██████	21	████████
4	████	10	█████	16	██████	22	████████
5	█████	11	██████	17	███████	23	██████████
6	██████	12	███████	18	████████	24	██████████

FIBER COUNT	BINDER UNIT (BU)	RING MARKINGS
144F	No Binder Unit	1-12 Ring Marking
288F (non-armored)	No Binder Unit	1-24 Ring Marking
288F (armored)	4 Binder Units	1-6 Ring Marking
432F	6 Binder Units	
576F	8 Binder Units	
864F	12 Binder Units	
1152F	8 Binder Units	
1728F	12 Binder Units	1-12 Ring Marking

WTC Design and Specification (Fiber Group ID)

Fiber Group ID – 200 um:

1-12 Ring Marks: 1728 – 3456 Fiber

144F SWR Binder Groups

1-24 Ring Marks: 6,912 Fiber

288F SWR Binder Groups

Stripe Ring Fiber Identification—6,912

R NO.	STRIPE RING MARKING	R NO.	STRIPE RING MARKING	R NO.	STRIPE RING MARKING	R NO.	STRIPE RING MARKING
1	█	7	███	13	██████	19	██████████
2	██	8	████	14	████████	20	██████████
3	███	9	█████	15	██████	21	██████████
4	████	10	████	16	██████	22	██████████
5	█████	11	█████	17	██████	23	██████████
6	█████	12	█████	18	██████	24	██████████

FIBER COUNT		BINDER UNIT (BU)												RING MARKINGS
1,728F	12 Binder Units	1	2	3	4	5	6	7	8	9	10	11	12	
3,456F	24 Binder Units	1	2	3	4	5	6	7	8	9	10	11	12	1-12 Ring Marking
		13	14	15	16	17	18	19	20	21	22	23	24	1-12 Ring Marking
6,912F	24 Binder Units	1	2	3	4	5	6	7	8	9	10	11	12	1-24 Ring Marking
		13	14	15	16	17	18	19	20	21	22	23	24	1-24 Ring Marking

*For binder units 13-24, the second binder unit is clear

WTC Design and Specification (Fiber Charts)

WTC Optical Fiber Charts

144 – 864 Fiber SWR Shown

BINDER NO.	COLOR	RIBBON NO.	FIBER COLOR											
			F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
			BLUE	ORANGE	GREEN	BROWN	SLATE	WHITE	RED	CLEAR	YELLOW	VIOLET	ROSE	AQUA
BINDER #1		SWR 1	1	2	3	4	5	6	7	8	9	10	11	12
		SWR 2	13	14	15	16	17	18	19	20	21	22	23	24
		SWR 3	25	26	27	28	29	30	31	32	33	34	35	36
		SWR 4	37	38	39	40	41	42	43	44	45	46	47	48
		SWR 5	49	50	51	52	53	54	55	56	57	58	59	60
		SWR 6	61	62	63	64	65	66	67	68	69	70	71	72
		SWR 7	73	74	75	76	77	78	79	80	81	82	83	84
		SWR 8	85	86	87	88	89	90	91	92	93	94	95	96
		SWR 9	97	98	99	100	101	102	103	104	105	106	107	108
		SWR 10	109	110	111	112	113	114	115	116	117	118	119	120
		SWR 11	121	122	123	124	125	126	127	128	129	130	131	132
		SWR 12	133	134	135	136	137	138	139	140	141	142	143	144
BINDER #2		SWR 1	145	146	147	148	149	150	151	152	153	154	155	156
		SWR 2	157	158	159	160	161	162	163	164	165	166	167	168
		SWR 3	169	170	171	172	173	174	175	176	177	178	179	180
		SWR 4	181	182	183	184	185	186	187	188	189	190	191	192
		SWR 5	193	194	195	196	197	198	199	200	201	202	203	204
		SWR 6	205	206	207	208	209	210	211	212	213	214	215	216
		SWR 7	217	218	219	220	221	222	223	224	225	226	227	228
		SWR 8	229	230	231	232	233	234	235	236	237	238	239	240
		SWR 9	241	242	243	244	245	246	247	248	249	250	251	252
		SWR 10	253	254	255	256	257	258	259	260	261	262	263	264
		SWR 11	265	266	267	268	269	270	271	272	273	274	275	276
		SWR 12	277	278	279	280	281	282	283	284	285	286	287	288
BINDER #3		SWR 1	289	290	291	292	293	294	295	296	297	298	299	300
		SWR 2	301	302	303	304	305	306	307	308	309	310	311	312
		SWR 3	313	314	315	316	317	318	319	320	321	322	323	324
		SWR 4	325	326	327	328	329	330	331	332	333	334	335	336
		SWR 5	337	338	339	340	341	342	343	344	345	346	347	348
		SWR 6	349	350	351	352	353	354	355	356	357	358	359	360
		SWR 7	361	362	363	364	365	366	367	368	369	370	371	372
		SWR 8	373	374	375	376	377	378	379	380	381	382	383	384
		SWR 9	385	386	387	388	389	390	391	392	393	394	395	396
		SWR 10	397	398	399	400	401	402	403	404	405	406	407	408
		SWR 11	409	410	411	412	413	414	415	416	417	418	419	420
		SWR 12	421	422	423	424	425	426	427	428	429	430	431	432

BINDER NO.	COLOR	RIBBON NO.	FIBER COLOR											
			F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
			BLUE	ORANGE	GREEN	BROWN	SLATE	WHITE	RED	CLEAR	YELLOW	VIOLET	ROSE	AQUA
BINDER #4		SWR 1	433	434	435	436	437	438	439	440	441	442	443	444
		SWR 2	445	446	447	448	449	450	451	452	453	454	455	456
		SWR 3	457	458	459	460	461	462	463	464	465	466	467	468
		SWR 4	469	470	471	472	473	474	475	476	477	478	479	480
		SWR 5	481	482	483	484	485	486	487	488	489	490	491	492
		SWR 6	493	494	495	496	497	498	499	500	501	502	503	504
		SWR 7	505	506	507	508	509	510	511	512	513	514	515	516
		SWR 8	517	518	519	520	521	522	523	524	525	526	527	528
		SWR 9	529	530	531	532	533	534	535	536	537	538	539	540
		SWR 10	541	542	543	544	545	546	547	548	549	550	551	552
		SWR 11	553	554	555	556	557	558	559	560	561	562	563	564
		SWR 12	565	566	567	568	569	570	571	572	573	574	575	576
BINDER #5		SWR 1	577	578	579	580	581	582	583	584	585	586	587	588
		SWR 2	589	590	591	592	593	594	595	596	597	598	599	600
		SWR 3	601	602	603	604	605	606	607	608	609	610	611	612
		SWR 4	613	614	615	616	617	618	619	620	621	622	623	624
		SWR 5	625	626	627	628	629	630	631	632	633	634	635	636
		SWR 6	637	638	639	640	641	642	643	644	645	646	647	648
		SWR 7	649	650	651	652	653	654	655	656	657	658	659	660
		SWR 8	661	662	663	664	665	666	667	668	669	670	671	672
		SWR 9	673	674	675	676	677	678	679	680	681	682	683	684
		SWR 10	685	686	687	688	689	690	691	692	693	694	695	696
		SWR 11	697	698	699	700	701	702	703	704	705	706	707	708
		SWR 12	709	710	711	712	713	714	715	716	717	718	719	720
BINDER #6		SWR 1	721	722	723	724	725	726	727	728	729	730	731	732
		SWR 2	733	734	735	736	737	738	739	740	741	742	743	744
		SWR 3	745	746	747	748	749	750	751	752	753	754	755	756
		SWR 4	757	758	759	760	761	762	763	764	765	766	767	768
		SWR 5	769	770	771	772	773	774	775	776	777	778	779	780
		SWR 6	781	782	783	784	785	786	787	788	789	790	791	792
		SWR 7	793	794	795	796	797	798	799	800	801	802	803	804
		SWR 8	805	806	807	808	809	810	811	812	813	814	815	816
		SWR 9	817	818	819	820	821	822	823	824	825	826	827	828
		SWR 10	829	830	831	832	833	834	835	836	837	838	839	840
		SWR 11	841	842	843	844	845	846	847	848	849	850	851	852
		SWR 12	853	854	855	856	857	858	859	860	861	862	863	864

WTC 144 – 6,912 Fiber Identification Chart Link Location: Page 120

WTC 250 μ m Design and Specification (Online)

250 μ m Wrapping Tube Cable (WTC) with SpiderWeb Ribbon® (SWR®)

Mechanical Data

DESCRIPTION	FIBER COUNT	BINDER UNIT	NOMINAL DIAMETER	WEIGHT	SHORT TERM / INSTALLATION		LONG TERM / STORAGE / STATIC	
			inches (mm)	lbs / 1,000 ft (kg/km)	MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)	MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)
NON-ARMORED								
LWSE-144-9-C-144-1-00N1D	144	1 X 144F	0.41 (10.5)	57 (85)	607 (2700)	9 (229)	182 (810)	6 (158)
LWSE-288-9-C-288-1-00N1D	288	1 X 288F	0.47 (12.0)	71 (105)	607 (2700)	10 (254)	182 (810)	7 (180)
LWSE-432-9-C-72-6-00N1D	432	6 X 72F	0.53 (13.5)	91 (135)	607 (2700)	11 (270)	182 (810)	8 (203)
LWSE-576-9-C-72-8-00N1D	576	8 X 72F	0.59 (15.0)	111 (165)	607 (2700)	12 (300)	182 (810)	9 (225)
LWSE-864-9-C-72-12-00N1D	864	12 X 72F	0.69 (17.5)	145 (215)	607 (2700)	14 (350)	182 (810)	11 (279)
LWSE-1152-K-C-144-8-00N1D	1152	8 X 144F	0.73 (18.5)	161 (240)	607 (2700)	15 (370)	182 (810)	11 (279)
LWSE-1728-K-C-144-12-00N1D	1728	12 X 144F	0.91 (23.0)	242 (360)	607 (2700)	18 (460)	182 (810)	14 (345)
OSP ARMORED*								
LWSE-144-9-C-144-1-10S1D	144	1 X 144F	0.63 (16.0)	148 (220)	607 (2700)	13 (320)	182 (810)	10 (254)
LWSE-288-9-C-72-4-10S1D	288	4 X 72F	0.69 (17.5)	172 (255)	607 (2700)	14 (350)	182 (810)	11 (279)
LWSE-432-9-C-72-6-10S1D	432	6 X 72F	0.75 (19.0)	202 (300)	607 (2700)	15 (380)	182 (810)	11 (285)
LWSE-576-9-C-72-8-10S1D	576	8 X 72F	0.81 (20.5)	235 (350)	607 (2700)	16 (410)	182 (810)	12 (308)
LWSE-864-9-C-72-12-10S1D	864	12 X 72F	0.91 (23.0)	286 (425)	607 (2700)	18 (460)	182 (810)	14 (345)
LWSE-1728-K-C-144-12-10S1D	1728*	12 X 144F	1.14 (29.0)	410 (610)	607 (2700)	23 (580)	182 (810)	17 (435)

* NOTE: Modified temperature performance

Note: Always reference the latest AFL updates at:
[Wrapping Tube Cable WTC with SWR \(aflglobal.com\)](http://WrappingTubeCableWTCwithSWR.aflglobal.com)

WTC 200 μ m Design and Specification (Online)

200 μ m Wrapping Tube Cable (WTC) with SpiderWeb Ribbon® (SWR®)

Mechanical Data—Non-Armored

DESCRIPTION	FIBER COUNT	BINDER UNIT	NOMINAL DIAMETER	WEIGHT lbs / 1,000 ft (kg/km)	SHORT TERM / INSTALLATION		LONG TERM / STORAGE / STATIC	
			inches (mm)		MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)	MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)
LWSE-1728-BE-C-144-12-00N1D	1,728	12 X 144F	0.85 (21.5)	202 (300)	607 (2700)	17 (432)	182 (810)	13 (330)
LWSE-3456-BE-C-144-24-00N1D	3,456	24 X 144F	1.04 (26.5)	292 (435)	607 (2700)	21 (533)	182 (810)	16 (406)
LWSE-6912-BE-C-288-24-00N1D	6,912	24 X 288F	1.38 (35.0)	514 (765)	607 (2700)	28 (711)	182 (810)	21 (533)

Note: Always reference the latest AFL updates at:
[200 \$\mu\$ m Wrapping Tube Cable WTC with SWR \(aflglobal.com\)](http://aflglobal.com)

WTC Design and Specification (GR-20)

WTC – GR-20 Specifications:

Qualifications

GOVERNING BODY	STANDARD CODE	COMPONENT
Telcordia	GR-20	Fiber Optic Cable

Contact AFL for further details.

TEMPERATURE RANGE

OPERATION	-40°F to +158°F (-40°C to +70°C)
STORAGE	-40°F to +158°F (-40°C to +70°C)
INSTALLATION	-22°F to +140°F (-30°C to +60°C)



Wrapping Tube Cable (WTC) Cable Reel Guidelines

Cable Reel Guidelines (Reel Handling)

Handling of Cable Reels:

When moving cable reels, care should be taken to ensure that **material handling equipment does not encounter any contact with the cable surface.**

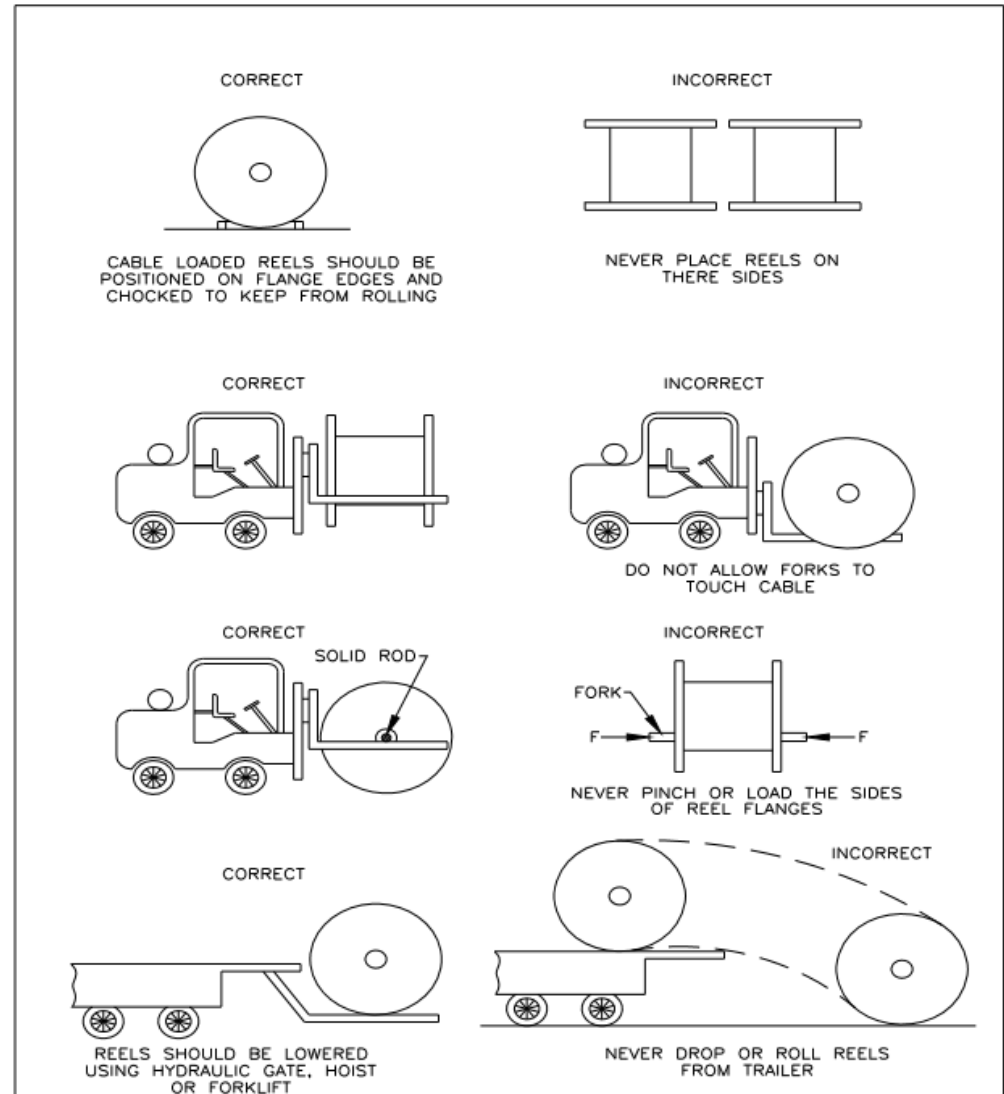
UNDER NO CIRCUMSTANCE SHOULD CABLE REELS BE DROPPED FROM ANY HEIGHT OR ALLOWED TO ROLL UNCONTROLLED

Whenever possible, cable reels should be moved or lifted using the following methods:

- For **cranes, booms, or other overhead lifting equipment**, a heavy steel arbor or suitable heavy rod or pipe should be inserted through the reel hubs allowing the cable reel to be lifted by **slings utilizing spreader bars or a lifting yoke**. This method will minimize the sling pressure against the reel flanges and reduce the risk of the reel tipping.
- When **lifting reels with fork truck type equipment**, reels should be lifted facing the flange with fork blades that are long enough to reach beneath both flanges. This method will ensure that the lift pressure is equally distributed on both flanges. Whenever possible, reels should be lifted with the flange that includes the **cable “test tail” facing away from the fork truck**. Do NOT lift the reel with the fork blades resting on the cable.

Cable Reel Guidelines (Reel Handling)

- Cable reels may be **moved short distances by rolling**, keeping in mind that reels should always be **rolled in the direction indicated by the arrows painted on the outside of the reel flanges**. This will prevent **loosening of the cable on the reel** which may cause **problems during installation**.
- The path over which the cable reels are to be **rolled must be clear of debris** that **might damage** the cable.
- Cable reels that are unloaded **down ramps** should be rolled, in a **controlled manner**, in the **direction** of the **arrows** painted on the reel flanges. The Ramp must be in a **gradual incline**, spaced parallel, and wide enough to **ensure contact** continuously with **both reel flanges** during unloading.



Cable Reel Guidelines (Reel Inspection)

Reel Inspection:

- When the cable reels are **initially received**, it is a common practice to **examine each reel** for potential **shipping damage**.
- During the **inspection**, the reel should be **checked for imperfections** including **nails** and **broken flanges**, which may cause problems during the installation process.
- If there are signs of **damage** or **repackaging** of the outer **protective wrap**, it is recommended that the wrap **be removed**, and the **cable surface** be **examined**.
- If the inspection is performed, replace the **protective wrap** around the cable. This is crucial to keep the cable **clean** and help **prevent future damage**.

Long Term:

- If the cable's wooden reel is pre-planned to be stored outside for **more than 6 months**, a **steel reel** should be **considered**.
- If the cable is on a **wooden reel** and has been in the weather for **longer than 6 months**, it may be necessary to **tighten the flange bolts** prior to installing the cable.

Wrapping Tube Cable (WTC) Pulling Installation Parameters

Installation Parameters (250 μ m - Cable Ratings)

Short Term (During Installation) – Dynamic
Long Term (After Installation) – Static

Note: Always reference the latest AFL updates at:
[Wrapping Tube Cable WTC with SWR \(aflglobal.com\)](http://aflglobal.com)

250 μ m Wrapping Tube Cable (WTC) with SpiderWeb Ribbon® (SWR®)

Mechanical Data

DESCRIPTION	FIBER COUNT	BINDER UNIT	NOMINAL DIAMETER	WEIGHT	SHORT TERM / INSTALLATION		LONG TERM / STORAGE / STATIC	
			inches (mm)	lbs / 1,000 ft (kg/km)	MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)	MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)
NON-ARMORED								
LWSE-144-9-C-144-1-00N1D	144	1 X 144F	0.41 (10.5)	57 (85)	607 (2700)	9 (229)	182 (810)	6 (158)
LWSE-288-9-C-288-1-00N1D	288	1 X 288F	0.47 (12.0)	71 (105)	607 (2700)	10 (254)	182 (810)	7 (180)
LWSE-432-9-C-72-6-00N1D	432	6 X 72F	0.53 (13.5)	91 (135)	607 (2700)	11 (270)	182 (810)	8 (203)
LWSE-576-9-C-72-8-00N1D	576	8 X 72F	0.59 (15.0)	111 (165)	607 (2700)	12 (300)	182 (810)	9 (225)
LWSE-864-9-C-72-12-00N1D	864	12 X 72F	0.69 (17.5)	145 (215)	607 (2700)	14 (350)	182 (810)	11 (279)
LWSE-1152-K-C-144-8-00N1D	1152	8 X 144F	0.73 (18.5)	161 (240)	607 (2700)	15 (370)	182 (810)	11 (279)
LWSE-1728-K-C-144-12-00N1D	1728	12 X 144F	0.91 (23.0)	242 (360)	607 (2700)	18 (460)	182 (810)	14 (345)
OSP ARMORED*								
LWSE-144-9-C-144-1-10S1D	144	1 X 144F	0.63 (16.0)	148 (220)	607 (2700)	13 (320)	182 (810)	10 (254)
LWSE-288-9-C-72-4-10S1D	288	4 X 72F	0.69 (17.5)	172 (255)	607 (2700)	14 (350)	182 (810)	11 (279)
LWSE-432-9-C-72-6-10S1D	432	6 X 72F	0.75 (19.0)	202 (300)	607 (2700)	15 (380)	182 (810)	11 (285)
LWSE-576-9-C-72-8-10S1D	576	8 X 72F	0.81 (20.5)	235 (350)	607 (2700)	16 (410)	182 (810)	12 (308)
LWSE-864-9-C-72-12-10S1D	864	12 X 72F	0.91 (23.0)	286 (425)	607 (2700)	18 (460)	182 (810)	14 (345)
LWSE-1728-K-C-144-12-10S1D	1728*	12 X 144F	1.14 (29.0)	410 (610)	607 (2700)	23 (580)	182 (810)	17 (435)

* NOTE: Modified temperature performance

Installation Parameters (200 μ m - Cable Ratings)

Short Term (During Installation) – Dynamic
Long Term (After Installation) – Static

Note: Always reference the latest AFL updates at:
[200 \$\mu\$ m Wrapping Tube Cable WTC with SWR \(aflglobal.com\)](http://aflglobal.com)

200 μ m Wrapping Tube Cable (WTC) with SpiderWeb Ribbon® (SWR®)

Mechanical Data—Non-Armored

DESCRIPTION	FIBER COUNT	BINDER UNIT	NOMINAL DIAMETER	WEIGHT lbs / 1,000 ft (kg/km)	SHORT TERM / INSTALLATION		LONG TERM / STORAGE / STATIC	
			inches (mm)		MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)	MAX TENSILE LOAD lbs (N)	MIN BEND RADIUS inches (mm)
LWSE-1728-BE-C-144-12-00N1D	1,728	12 X 144F	0.85 (21.5)	202 (300)	607 (2700)	17 (432)	182 (810)	13 (330)
LWSE-3456-BE-C-144-24-00N1D	3,456	24 X 144F	1.04 (26.5)	292 (435)	607 (2700)	21 (533)	182 (810)	16 (406)
LWSE-6912-BE-C-288-24-00N1D	6,912	24 X 288F	1.38 (35.0)	514 (765)	607 (2700)	28 (711)	182 (810)	21 (533)

Installation Parameters (Tensile Load)

AFL WTC:

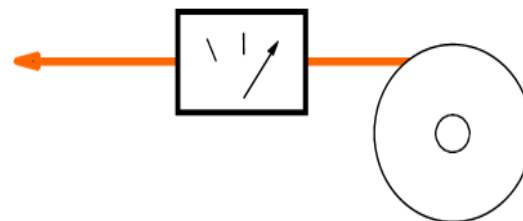
- **Never Exceed the Cables Rated Tensile Load.**
- Each cable has 2 published strengths:

Short Term / Installation / Dynamic:

During installation and under tension.

Long Term / Storage / Static:

In operation and the cable is in a relaxed state.



Outside Plant Cable

Short Term - **Maximum Tensile Load - 600 lbs.**

Long Term - **Maximum Tensile Load - 180 lbs.**

Installation Parameters (Minimum Bend Radius)

AFL WTC Minimum Bend Radius:

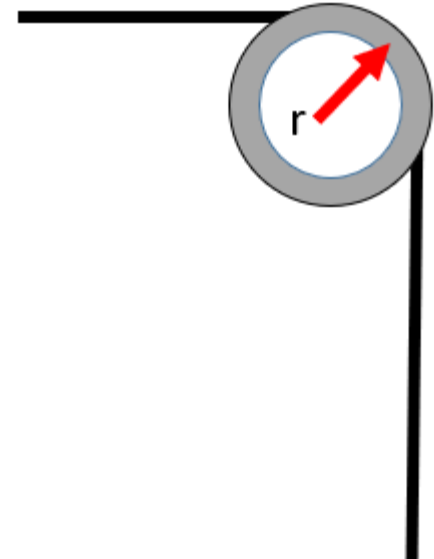
- **Never Exceed** the **Cables Minimum Bend Radius**.
- Each cable has 2 published Bend Radii:

Short Term / Installation / Dynamic:

During installation and under tension.

Long Term / Storage / Static:

In operation and the cable is in a relaxed state.



Outside Plant Cable

Short Term - **Minimum** Bend Radius - **20 X** Cable Diameter

Long Term - **Minimum** Bend Radius - **15 X** Cable Diameter

Radius x 2 = Diameter

Installation Parameters (Sheave / Capstan Wheel)

AFL WTC Sheave / Capstan Wheel:

- **Rated** for the **maximum installation tension** of (600 lbs.)
- **Surface meets or exceeds** the **minimum bend radius: 20X Cable Diameter**
- Measured at the **Root Diameter** (The lowest part of the wheel / Actual contact surface.)
- **Good** overall working condition. No **stiff** or **frozen** bearings.
- **Clean smooth surface or lining** and **without** damage or burrs.
- **Anchored / Secured** during the installation.



Installation Parameters (Equipment)

AFL WTC:

- **Rated for Max Pull** tension (600+ lbs.)
- **Max** pull rate of **100' to 150'** per minute.
- **Anchor** and **Chock** during pull.
- Capable of **Monitoring Tension**.
- Equipped with a **Clutch Tension Limiter**.



GMP Sidewinder Fiber Optic Puller Shown:

<https://www.gmptools.com/sidewinder-fiber-optic-puller/>

Installation Parameters (Equipment)

Typical Cable Pulling Machines:

Cable pulling machines are available through several different manufactures. Based on cables size and the overall distance that is required, **most contractors will use the same equipment that they use for any traditional pulled cable.**

The models below are typical for a WTC installation:



GMP SideWinder



Condux 08675675



GMP

CONDUX – Fiber Optic Cable Puller (Video): https://www.youtube.com/watch?v=B7Imf_eJGlo&app=desktop

Wrapping Tube Cable (WTC) Proofing the Pathway (Pre-Pulling)

Proofing the Pathway (Pre-Pulling)

Proofing the Pathway: (Pre-Pulling)

- **Blow a foam lube spreader or pull a foam line carrier** through the duct before installing the cable to make sure there is **no water or debris within the duct**.
- If necessary, **flush the duct** with water to clear **dirt and debris**.
- Perform a duct **conformance test** using a **proper sized proofing ball or duct mandrel**. Proofing the duct is used to **check the joint integrity and confirm the ovality** of the inner diameter.
- Always **pressure test the duct** per standard procedures to **verify** the integrity of the **duct and couplers**.
- If the **proofing test fails**, **repair the issues** that were found and **re-proof** the duct.
- **Lube the duct before and throughout** the run using an **approved industry standard lubricant** per standard installation methods..



Water in Duct

Lube Requirements: (Pulling Lube)

Must be **compatible** with the cable **jacket type**, good **coating** and **cling rates**, **MSDS** safe, **no environmental impact**, easy **clean up**, **slow dry** factor, **easy dispensing** system, available in **cold weather formula**.

Proofing the Pathway (Pre-Pulling)

Proofing Examples:

Foam Line Carriers: (1)

<https://www.gmptools.com/line-carriers/>

Foam Lube Spreader: (2)

<https://www.comstarsupply.com/1-1-4-foam-lube-spreader.html>

Duct Proofing Balls: (3)

<https://www.mcmaster.com/acetal-homopolymer-balls>

Duct Mandrels: (4)

<https://www.stanlay.com/cable-installation-equipment/duct-integrity-testers/duct-mandrels.html>

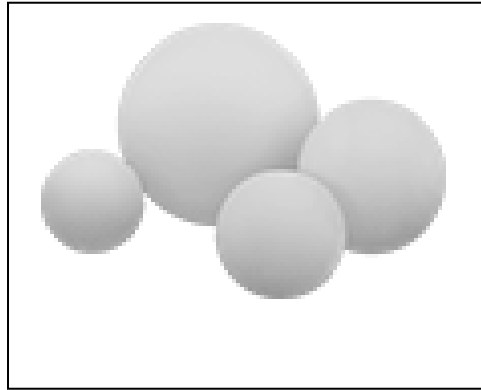
1.



2.



3.



4.



Wrapping Tube Cable (WTC) Key Points (Pre-Pulling)

Key Points for Pre-Pulling (Planning)

Key Points for Pre-Pulling: (Planning)

- Pre-identify the **cable route** and **sharp angle** prior to the installation.
- **If required, test** and verify the **optical integrity** of the reel being installed.
- Determine the **actual length** of the **span** to include all specified **storage** requirements.
- Verify that the **cable** meets the **required length** per the site location.
- Pre-identify a **safe** and **available working location** to place the installation equipment.
- Pre-identify a **safe** and **available working location** to perform the **Figure 8** or **Racetrack** coiling procedures.
- Pre-test the **communication devices** at each site prior to the installation.
- Have all **required permits** and **safety related equipment** together and available.

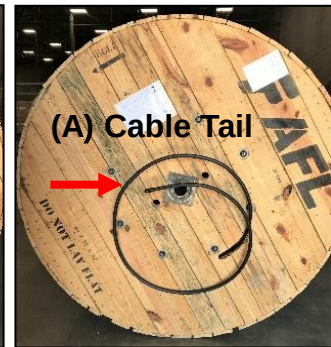
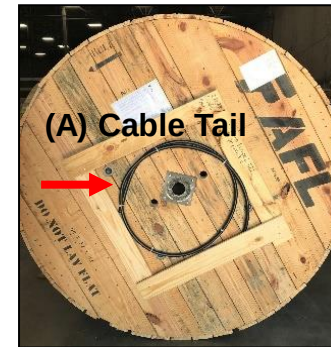


Key Points for Pre-Pulling (Setup)

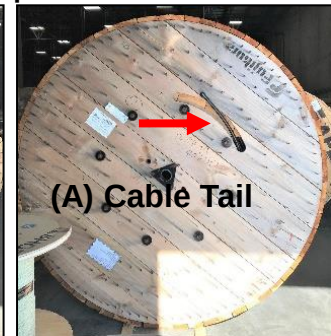
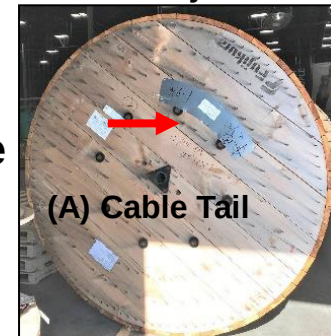
Key Points for Pre-Pulling: (Setup)

- Place the **cable** and **equipment** at the predetermined **location**.
- **Align the cable reel** with the **entry pathway** leaving enough room between the two for a **smooth transition (10–15 feet)**.
- **Remove** any **loops** of duct found inside the **storage areas** to reduce the friction during the installation.
- **Loosen** the “**inside cable tail**” (A) in order to allow the cable to move during the installation. **Failure** to perform this task may **loosen** the cable lay or potentially **damage** the cable. **Remove** any **excessive** length necessary from the tail (A) before or during the installation in order to prevent potential **entanglement issues**.
- Do not remove the **protective wrapping** from the reel until the cable is **ready to be installed**. This is crucial to keep the cable as clean as possible as well as to prevent any possible damage to the cable prior to the installation.
- Once the cable is ready to be installed, remove the protective covering and **re-inspect the reel** for **imperfections** which may cause issues during the installation process.

WTC: AFL USA



WTC: Fujikura Japan



Key Points for Pre-Pulling (Setup)

Key Points for Pre-Pulling: (Setup)

- Be sure to **maintain the cable wrap** or have a **tarp available** if needed to prevent the cable from being soiled during the installation. Dirt and debris entering the duct will **reduce the performance** of the installation.
- **Note:** This practice may need to be used throughout the **cable coiling process** as well (Fig.8 / Racetrack) depending on the condition of the surface where the cable is being stored.
- Review the cables **maximum tension** and **minimum bend radius specifications** based on the **latest AFL online specifications**.
- **Make sure** that the cable is **paying off** from the **top of the reel**.
- Check the **minimum bend radius** of the **cable entering and exiting** the duct.
- Make sure that the **proper pulling lubricant** was applied or is setup to be applied.
- Have the proper communication devices in place.



Key Points for Pre-Pulling (Installation)

Key Points for Pre-Pulling: (Installation)

- **Hand assists** in the **cable payoff** throughout the installation **if needed** in order to maintain a **low tension** on the cable.
- **Visually watch the cable** as it pays off the reel in order to avoid any loose cable from becoming entangled during the installation process.
- Try to maintain the **cable from touching the ground unprotected** between the reel and the duct.
- Have a **towel or wipe** available to remove any remaining **dirt or moisture** from the cable prior to entering the duct.



Wrapping Tube Cable (WTC) Duct Fill Ratio Consideration (Pulling Guidelines)

Non-Armored - Duct Fill Ratio (Pulling - 250 um)

Duct Fill Ratio Consideration / AFL WTC - 250 um SWR

Cable Pulling Conditions:

Conduit ID: Smooth Wall (Silicor) or Ribbed Wall

Span: Based on a 1,000 foot Maximum Span with Two (2) 90 Degree Turns.

Cable: One (1) Non-Armored Cable / Pulled with a Maximum Fill Ratio Target of 65%.

Acceptable

Borderline

Not Acceptable

AFL Specifications: 600 lb. Max Tension - 600 lb. Breakaway Swivel, Specified Lube

			Conduit Pipe Sizes - ID					
AFL WTC	Cable Diameter		0.75"	1.0"	1.25"	1.50"	1.75"	2.0"
Fiber Count	Inches / mm		19.05 mm	25.4 mm	31.75 mm	38.1 mm	44.45 mm	50.8 mm
144 Fiber	0.41	10.5	30.4%	17.1%	10.9%	7.6%	5.6%	4.3%
288 Fiber	0.47	12.0	39.7%	22.3%	14.3%	9.9%	7.3%	5.6%
432 fiber	0.53	13.5	50.2%	28.2%	18.1%	12.6%	9.2%	7.1%
576 Fibers	0.59	15.0	62.0%	34.9%	22.3%	15.5%	11.4%	8.7%
864 Fibers	0.69	17.5	84.4%	47.5%	30.4%	21.1%	15.5%	11.9%
1152 Fibers	0.73	18.5	94.3%	53.0%	34.0%	23.6%	17.3%	13.3%
1728 Fibers	0.91	23.0	145.8%	82.0%	52.5%	36.4%	26.8%	20.5%

Note: The fill ratio chart is calculated by using the cable and duct inner diameters only.

Any additional hardware (pulling grip, breakaway swivel etc.) needs to be added to the fill ratio calculation.

Fill Ratio Formula: (Ratio of the area of the cable verses the interior area of the conduit)

Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}}$ Where: Cable r = Cable Diameter/2 and Conduit Interior r = Conduit Interior Diameter/2 and $\pi = 3.14$

Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}} = \frac{\text{Cable } \pi r^2}{\text{Conduit } \pi r^2}$

Non-Armored - Duct Fill Ratio (Pulling - 200 um)

Duct Fill Ratio Consideration / AFL UHD WTC - 200 um SWR

Cable Pulling Conditions:

Conduit ID: Smooth Wall (Silicor) or Ribbed Wall

Span: Based on a 1,000 foot Maximum Span with Two (2) 90 Degree Turns.

Cable: One (1) Non-Armored Cable / Pulled with a Maximum Fill Ratio Target of 65%.

AFL Specifications: 600 lb. Max Tension - 600 lb. Breakaway Swivel, Specified Lube

Acceptable
Borderline
Not Acceptable

			Conduit Pipe Sizes - ID				
AFL WTC	Cable Diameter		1.0"	1.25"	1.50"	1.75"	2.0"
Fiber Count	Inches / mm		25.4 mm	31.75 mm	38.1 mm	44.45 mm	50.8 mm
1728 Fiber	0.85	21.5	71.6%	45.9%	31.8%	23.4%	17.9%
3456 Fiber	1.04	26.5	108.8%	69.7%	48.4%	35.6%	27.2%
6912 fiber	1.38	35.0	189.9%	121.5%	84.4%	62.1%	47.5%

Note: The fill ratio chart is calculated by using the cable and duct inner diameters only.

Any additional hardware (pulling grip, breakaway swivel, etc.) needs to be added to the fill ratio calculation.

Fill Ratio Formula: (Ratio of the area of the cable verses the interior area of the conduit)

Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}}$ Where: Cable r = Cable Diameter/2 and Conduit Interior r = Conduit Interior Diameter/2 and $\pi = 3.14$

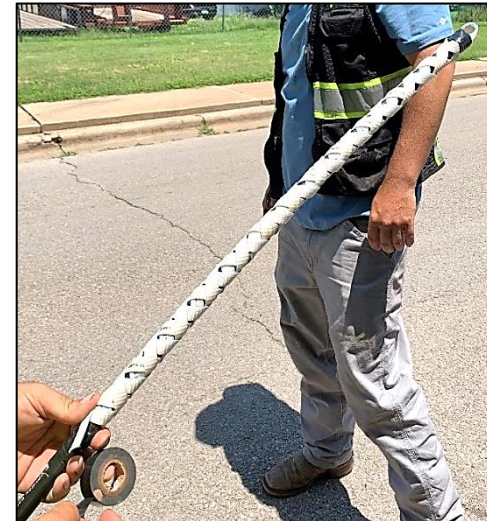
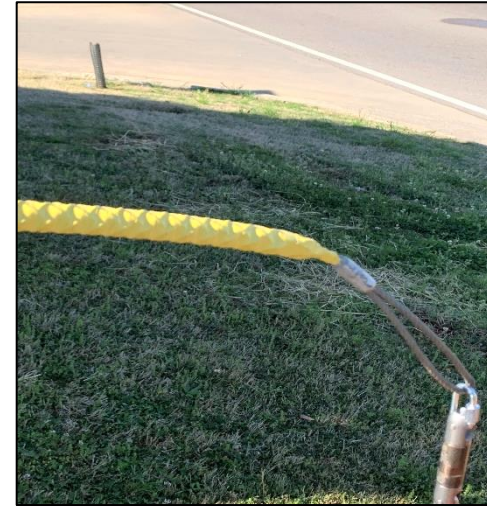
Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}} = \frac{\text{Cable } \pi * r^2}{\text{Conduit } \pi * r^2}$

Wrapping Tube Cable (WTC) Preparing the Cable End and Pulling Grip

Preparing the Cable End and Pulling Grip

Preparing the Cable End and Pulling Grip:

- When preparing the cable for pulling, be sure and cover the end with **vinyl tape** to prevent any potential **moisture** from **entering the inner portion** of the cable.
- Be sure that the **fiber optic pulling grip** or **mule tape method** exceeds the maximum tension of the installation. (Maximum Tension: 600 lbs.)
- Cover the pulling **grip with vinyl tape** to help reduce friction within the duct.
- Be sure to install a **600 lbs. double clevis breakaway swivel** on the end of the cable pulling eye.
- After the cable is pulled, **cut and remove approximately 15 feet** from the end of the cable. This process is performed to **remove any potential damage** to the optical fiber that may have **occurred during the installation**.



Wrapping Tube Cable (WTC) Break-Away Swivel and Pulling Grip

Break-Away Swivel and Pulling Grip

Swivel and Pulling Grip:

- Specific **FO** cable **pulling grip** (typically at **least 32 inches** long).
- Rated for **max pull tension** and **cable diameter range**.
- **2 Stage** – Double weave to single weave grip
- **Double Clevis Swivel**

Rated for Load
Break away at **600 lbs.** (2700N)

Swivel:

<https://products.condux.com/products.php?id=369>

Fiber Optic Pulling Grip:

<https://products.condux.com/products.php?id=317>



Wrapping Tube Cable (WTC) Cascade and Figure 8 Pulling Method

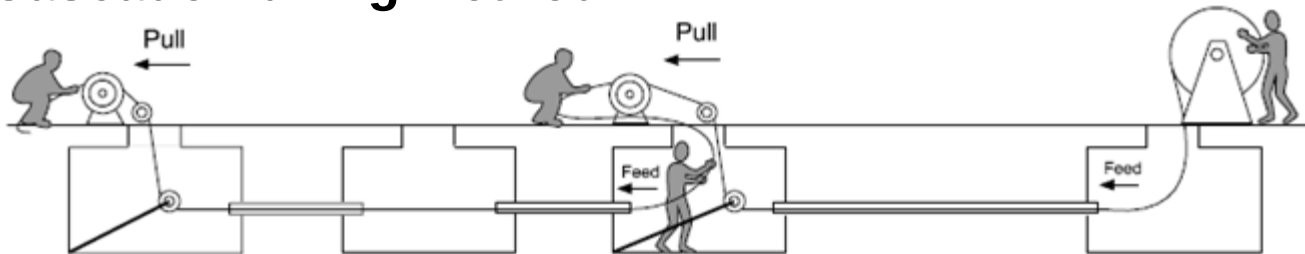
Cascade and Figure 8 Pulling Methods

Cascade Point Process:

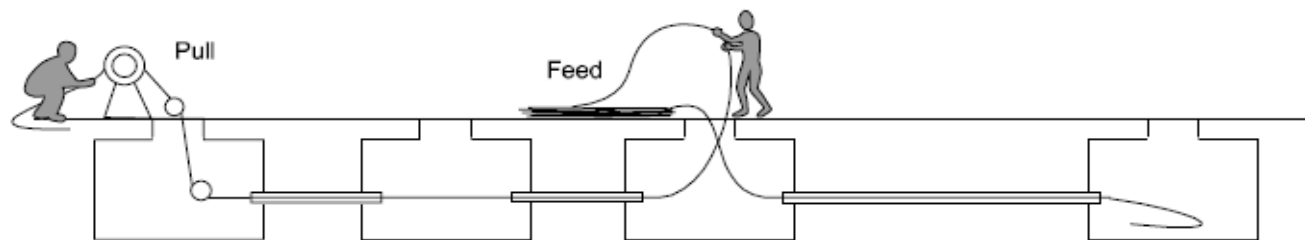
Below are two examples of the **cascade point process**. Each diagram shows a method **used to install the cable at a greater distance**.

Note: The detailed methods for both the Figure 8 and Racetrack procedures / video are located on pages 80 – 89 of this document.

Cascade Pulling Method:



Hand Figure 8 Pulling Method:



Wrapping Tube Cable (WTC) Key Points for Post Pulling

Key Points for Post Pulling

Post Pulling:

- **Leave enough cable** on each end to reach the intended application.
- Use one of the **AFL Recommended Methods to Coil the WTC**. **Never** use the **Fold and Flip** procedure to store the cable.
- **Secure all cable** against potential damage.
- **Identify and label** all cables for future applications.
- **Document the cable lengths, routing direction, fiber type and cable ID.**
- **Secure any open areas.**



Wrapping Tube Cable (WTC) Air Assisted Installation Parameters

Installation Parameters (Terminology - Jetting)

Cable Jetting Terminology: (Brief Overview)

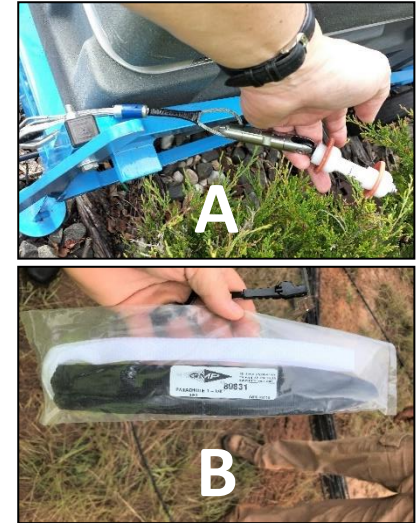
Cable **jetting** is the process of placing a fiber optic cable into an existing duct with the use of specialized installation equipment along with a **high-volume** air compressor.

This type of installation uses a **high pressure / low volume** source of air to help **pull** the cable through the duct with the use of a **Cable Carrier (A)** or **Parachute (B)**.

The installation process starts by properly attaching a **FO pulling grip, break-away swivel** and **cable carrier or parachute onto the end of the cable**. The cable is then properly placed into the equipment and tractor drive mechanism.

The air pressure is then applied which causes the **carrier or parachute** to build up **pressure** and form a **seal**. **Both** the speed of the **drive mechanism** and the amount of **air pressure** is then **independently adjusted** so that they both work together throughout the installation. The cables **overall pulling force** should **not exceed** the specified **maximum tensile rating**.

Note: The drive mechanism is designed to only **assist** the **air flow in moving the cable forward** and not to **push or force** the cable into the duct.

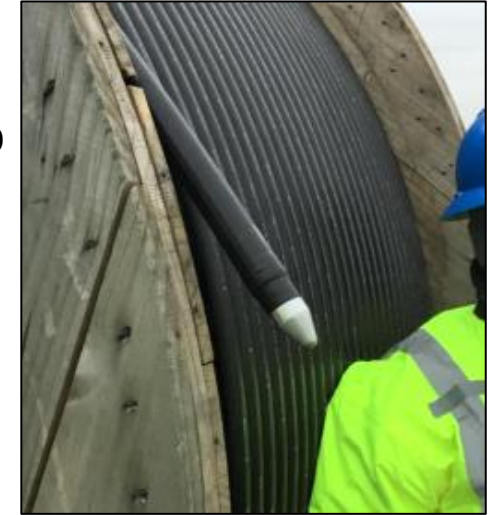


Installation Parameters (Terminology – Blowing)

Cable Blowing Terminology : (Brief Overview)

Cable **blowing** is the process of placing a fiber optic cable into an existing duct with the use of specialized installation equipment along with a **high-volume** air compressor.

This type of installation uses a **low pressure / high-volume** source of air to help “**float**” the cable through the duct and reduce its overall contact with the inner wall. **Cable tips** can also help to avoid **catch points** within the duct.



Typically, the installation process starts by properly placing the cable into the equipment and starting the tractor drive mechanism to install the cable **approximately 50 to 100 feet** into the duct **without air** and under **low tension**. This process is used to help prevent the cable from being pushed out once the initial air is applied.

The air pressure is then applied in **small increments** and allowed to stabilize prior to starting the drive mechanism. Once the stabilization is complete, both the speed of the **drive mechanism** and the amount of **air pressure** need to be **independently adjusted** so that they both work together throughout the installation.

Note: The drive mechanism is designed to only **assist** the **air flow in moving the cable forward** and not to **push or force** the cable into the duct.

Installation Parameters (Equipment)

Maximum Installation Pressure:

The **maximum installation pressure** is based on the **type of equipment selected** for each install. The type of **compressor** and its **maximum output** is also a factor in reaching the machines maximum potential.

Maximum Installation Rate (Speed):

The **maximum installation rate** is based on the **type of equipment selected** for each install. The **overall pathway** also plays a large part in the actual speed due the number of **angles** and **elevation changes**.

Examples of **typical installation machines** per the manufacture: (3-2021)

<u>Cable Jetting / Blowing Machines</u>	<u>Maximum Pressure</u>	<u>Maximum Speed</u>	<u>Cable Dia. Min. / Max.</u>	<u>Duct Dia. Min. / Max.</u>
Plumett MiniJet	16 bar	125 m/min.	4 - 16 mm	7 - 42 mm
Plumett Superjet Hydraulic	12 bar	90 m/min.	9 - 32 mm	20 - 63 mm
Condux Gulfstream 400	10-14 bar	90 m/min.	5.8 - 28.7 mm	10 - 51 mm
Condux Gulfstream 400 XL	7-10 bar	N/A	12 - 39 mm	32 - 63 mm
GMP Tornado	12 bar	90 m/min.	6 - 34 mm	12 - 63 mm
GMP Airstream	10-15 bar	80 m/min.	2.5 - 11 mm	5 - 18 mm

Installation Parameters (Equipment)

Air Assisted Installation Machines:

Air assisted installation machines are available through several different manufactures. Be sure to **use only** the **Installation Machines** that are rated for the **size of cable** that you're installing. Most contractors will use the **same equipment** that they use for any **traditional air blown cable**.

The models below are typical for an AFL WTC installation:



Plumett Superjet Hydraulic



Condux GulfStream 400 and XL



GMP Tornado

Plumett SuperJet: <https://www.plumettaz.com/en/product/superjet/H01>

Condux - Lightweight Fiber Optic Cable Blower (Video): <https://www.youtube.com/watch?v=8P8BqR-4T1Q>

Tornado Setup V2 (Video): https://www.youtube.com/watch?v=NDzJZ_K7rvE

Wrapping Tube Cable (WTC) Proofing the Pathway for Air Assisted Installations

Proofing the Pathway (Air Assisted Installations)

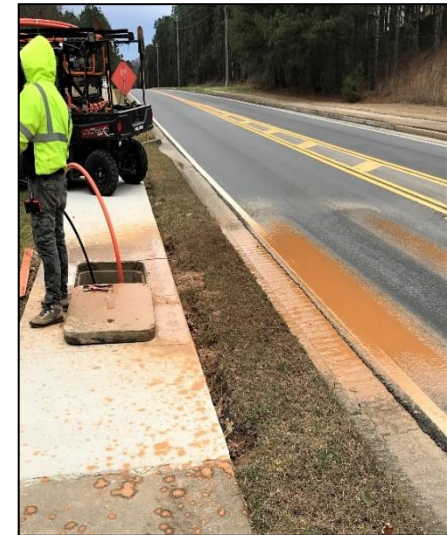
Proofing the Pathway: (Air Assisted Installations)

#1. Cleaning

- Before starting the cable installation, pressurize the conduit to flush out any **large amounts of water or debris**.
- If the debris is excessive, **flush the duct** with water to clear the pathway.
- **Blow a foam lube spreader or pull a foam line carrier** through the duct to clear out any remaining **water or debris**.

Note:

- A foam lube spreader or carrier **can not** detect **ovality** or **angle issues** within a duct. Failure to detect these issues prior to starting the installation can result in installation delays or cable damage.



Proofing the Pathway (Air Assisted Installations)

Proofing the Pathway: (Air Assisted Installations)

#2. Pressure Test:

- Pressure test the duct per standard procedures to **verify** the **integrity** of the **duct** and **couplers**.

Note:

- Loss of pressure within the pathway will reduce the installation performance.

#3. Conformance Test

- Perform a duct **conformance test** using a **proper sized proofing ball** or **duct mandrel**. The test is performed to identify any potential **damage**, **ovality**, **tight bends** or **elevation changes** that can affect the performance of the installation.
- If the **proofing test fails**, **repair** the **issues** that were found and **re-proof** the duct.
- The proper sized **ball** or **mandrel** should be **equal** to or **greater than** the cable being installed.



Proofing the Pathway (Air Assisted Installations)

Proofing Examples:

Foam Line Carriers: (1)

<https://www.gmptools.com/line-carriers/>

Foam Lube Spreader: (2)

<https://www.comstarsupply.com/1-1-4-foam-lube-spreader.html>

Duct Proofing Balls: (3)

<https://www.mcmaster.com/acetal-homopolymer-balls>

Duct Mandrels: (4)

<https://www.stanlay.com/cable-installation-equipment/duct-integrity-testers/duct-mandrels.html>

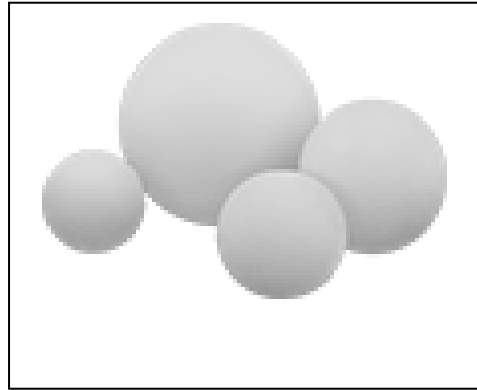
1.



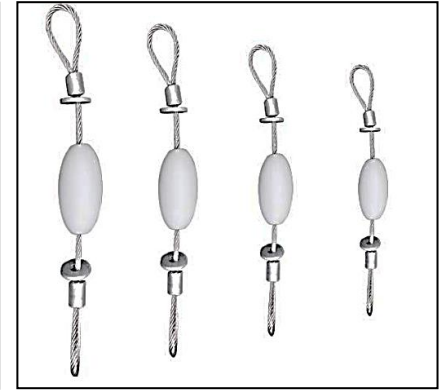
2.



3.



4.



Wrapping Tube Cable (WTC) Lubricating the Pathway for Air Assisted Installations

Lubricating the Pathway (Air Assisted Installations)

Lubricating the Pathway: (Air Assisted Installations)

- Each duct needs to be **cleaned**, **proof tested** and **lubricated** prior to starting the installation.
- Be sure to use an **approved industry standard lubricant** that is appropriate for the **season** your working in. Each lubricant should have instructions on **how much lube** should be use based on the **duct size and length of the pathway**.
- **Failure to sufficiently lubricant the duct** per the manufacture recommendation can result in **higher friction** and **unnecessary manual assistance** during the installation.

Lube Requirements: (Jetting / Blowing Lube)

Must be compatible with the cable jacket type, good coating and cling rates, MSDS safe, no environmental impact, easy clean up, slow dry factor, easy dispensing system.

Dura-line Hydralube: AT-500 Cable Pulling Lubricant
(Summer or Winter Grade) For Heavy Cables and Multiple Bends

https://www.mce1000.com/mainsource/59q/200-120050_spec.pdf

<https://www.duraline.com/globalassets/general/documents/hydra-lube/hydralube-at-500-winter-grade---na.pdf?v=497322>

https://www.duraline.com/-/ods/Resource/195/1230/Original/DL_Lubricant%20General%2012.2019.pdf?version=1591694662



Wrapping Tube Cable (WTC)

Key Points for Air Assisted Installations

Key Points for Air Assisted Installations (Planning)

Key Points for Air Assisted Installations: (Planning)

- Pre-identify the **cable route** and **sharp angle** prior to the installation. **NOTE:** Every 360° of turns = 1,000' of straight conduit.
- **If required, test** and verify the **optical integrity** of the reel being installed.
- Determine the **actual length** of the **span** to include all specified **storage** requirements.
- Verify that the **cable** meets the **required length** per the site location.
- Pre-identify a **safe** and **available working location** to place the installation equipment.
- Pre-identify a **safe** and **available working location** to perform the **Figure 8** or **Racetrack** coiling procedures.
- Pre-test the **communication devices** at each site prior to the installation.
- Have all **required permits** and **safety related equipment** together and available.

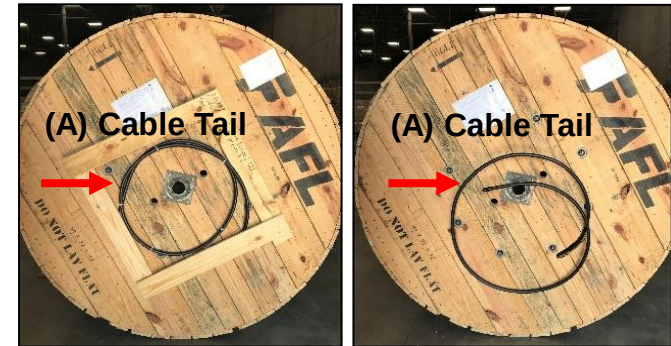


Key Points for Air Assisted Installations (Setup)

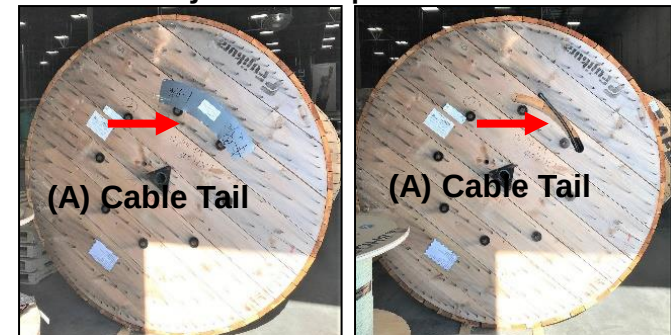
Key Points for Air Assisted Installations: (Setup)

- **Loosen** the “**inside cable tail**” (A) in order to allow the cable to move during the installation. **Failure** to perform this task may **loosen** the cable lay or potentially **damage** the cable. **Remove** any **excessive** length necessary from the tail (A) before or during the installation in order to prevent potential **entanglement issues**.
- Do not remove the **protective wrapping** from the reel until the cable is **ready to be installed**. This is crucial to keep the cable as clean as possible as well as to prevent any possible damage to the cable prior to the installation.
- Once the cable is ready to be installed, remove the protective covering and **re-inspect the reel** for **imperfections** which may cause issues during the installation process.

WTC: AFL USA



WTC: Fujikura Japan



Key Points for Air Assisted Installations (Setup)

Key Points for Air Assisted Installations: (Setup)

- Be sure to **maintain** the **cable wrap** and have a **tarp available** if needed to prevent the cable from being soiled during the installation. **Dirt and debris** entering the **duct** and or **machine** **will reduce the performance** of the installation.
- **Note:** This practice may need to be used throughout the **cable coiling process** as well (Fig.8 / Racetrack) depending on the condition of the surface where the cable is being stored.
- **Make sure** that the cable is **paying off** from the **top of the reel**.
- **Align the cable reel** with the **entry pathway** leaving enough room between the two for a **smooth transition (10–20 feet)**.



Key Points for Air Assisted Installations (Setup)

Key Points for Air Assisted Installations: (Setup)

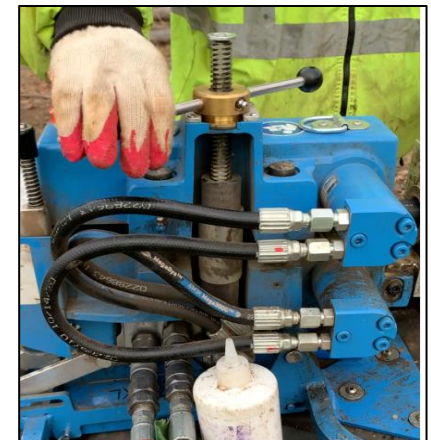
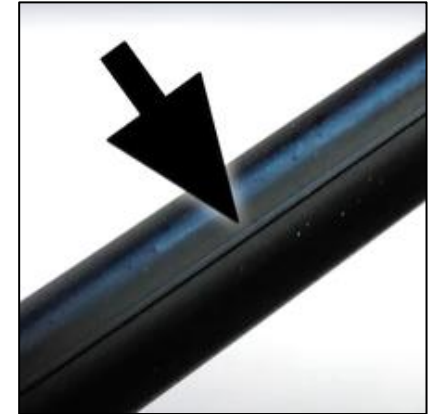
- Place the blowing equipment **closer to the entrance** of the manhole or vault. This process will help eliminate unnecessary angles.
- Review the cables **Maximum Tension** and **Minimum Bend Radius** ratings based on the **latest AFL online specifications**.
- Pre-check what the **Minimum Bend Radius** will be **entering** and **exiting** the duct.
- Make sure the crew has the **proper equipment** to prevent the cable from exceeding the **Minimum Bend Radius**.



Key Points for Air Assisted Installations (Installation)

Key Points for Air Assisted Installations: (Installation)

- Place the **WTC** into the **installation machine** with the cable **ridge up** (natural bend of the cable). This will allow the cable to **move freely** with the **flow of the pathway**.
- **Never over tighten** the **installation machine drive belt** more than what is **recommended** by the **equipment manufacture**.
- **Hand assists** the **payoff reel (if not powered)** to reduce the tension on the installation machine and to prevent the cable from exceeding the **max installation tension** of 600 lbs.
- Always have someone **maintain an onsite visual watch** of the **reel** to prevent potential slack, loops and snags. The process is especially important at **both flange locations**.



Key Points for Air Assisted Installations (Tips)

Key Points for Air Assisted Installations: (Tips)

- Try to maintain the **cable from touching the ground unprotected** between the reel and the duct.
- Have a **towel or wipe** available to remove any remaining **dirt or moisture** from the cable prior to entering the duct.
- During **High Humidity** installations, an **aftercooler** may be necessary to cool the compressed air and remove water droplets prior to entering the cable installation machine.
- If you have a **static build up** within the **duct**, try wiping the cable with an **anti-static dryer sheet** before the cable enters the duct.



Important:

- **Maintain installation records.** If a cable is showing signs of **damage**, make a note of the **installer, date, location, issue, and foot markings**. Additional **photographs** may be helpful in determining where the issue began and how to prevent it from reoccurring in the future.

Wrapping Tube Cable (WTC) Duct Fill Ratio Consideration (Air Assisted Installations)

Non-Armored - Duct Fill Ratio (250 um)

Duct Fill Ratio Consideration / AFL WTC - 250 um SWR

Cable Air Assisted Conditions:

Conduit ID: Smooth Wall (Silicor) or Ribbed Wall

Air: Blowing Clean Dry Air

Span: Based on a 2,000 foot Maximum Span with Two (2) 90 Degree Turns.

Cable: One (1) Non-Armored Cable / Installed with a Maximum Fill Ratio Target of 75%.

Acceptable
Borderline
Not Acceptable

AFL Specifications: 600 lb. Max Tension - 600 lb. Breakaway Swivel, Specified Lube

			Conduit Pipe Sizes - ID					
AFL WTC	Cable Diameter		0.75"	1.0"	1.25"	1.50"	1.75"	2.0"
Fiber Count	Inches / mm		19.05 mm	25.4 mm	31.75 mm	38.1 mm	44.45 mm	50.8 mm
144 Fiber	0.41	10.5	30.4%	17.1%	10.9%	7.6%	5.6%	4.3%
288 Fiber	0.47	12.0	39.7%	22.3%	14.3%	9.9%	7.3%	5.6%
432 fiber	0.53	13.5	50.2%	28.2%	18.1%	12.6%	9.2%	7.1%
576 Fibers	0.59	15.0	62.0%	34.9%	22.3%	15.5%	11.4%	8.7%
864 Fibers	0.69	17.5	84.4%	47.5%	30.4%	21.1%	15.5%	11.9%
1152 Fibers	0.73	18.5	94.3%	53.0%	34.0%	23.6%	17.3%	13.3%
1728 Fibers	0.91	23.0	145.8%	82.0%	52.5%	36.4%	26.8%	20.5%

Note: The fill ratio chart is calculated by using the cable and duct inner diameters only.

Any additional hardware (pulling grip, breakaway swivel, carriers etc.) needs to be added to the fill ratio calculation.

Fill Ratio Formula: (Ratio of the area of the cable verses the interior area of the conduit)

Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}}$ Where: Cable r = Cable Diameter/2 and Conduit Interior r = Conduit Interior Diameter/2 and $\pi = 3.14$

Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}} = \frac{\text{Cable } \pi * r^2}{\text{Conduit } \pi * r^2}$

Non-Armored - Duct Fill Ratio (200 um)

Duct Fill Ratio Consideration / AFL UHD WTC - 200 um SWR

Cable Air Assisted Conditions:

Conduit ID: Smooth Wall (Silicor) or Ribbed Wall

Air: Blowing Clean Dry Air

Span: Based on a 2,000 foot Maximum Span with Two (2) 90 Degree Turns.

Cable: One (1) Non-Armored Cable / Installed with a Maximum Fill Ratio Target of 75%.

Acceptable
Borderline
Not Acceptable

AFL Specifications: 600 lb. Max Tension - 600 lb. Breakaway Swivel, Specified Lube

			Conduit Pipe Sizes - ID				
AFL WTC	Cable Diameter		1.0"	1.25"	1.50"	1.75"	2.0"
Fiber Count	Inches / mm		25.4 mm	31.75 mm	38.1 mm	44.45 mm	50.8 mm
1728 Fiber	0.85	21.5	71.6%	45.9%	31.8%	23.4%	17.9%
3456 Fiber	1.04	26.5	108.8%	69.7%	48.4%	35.6%	27.2%
6912 fiber	1.38	35.0	189.9%	121.5%	84.4%	62.1%	47.5%

Note: The fill ratio chart is calculated by using the cable and duct inner diameters only.

Any additional hardware (pulling grip, breakaway swivel, carriers etc.) needs to be added to the fill ratio calculation.

Fill Ratio Formula: (Ratio of the area of the cable verses the interior area of the conduit)

Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}}$ Where: Cable r = Cable Diameter/2 and Conduit Interior r = Conduit Interior Diameter/2 and $\pi = 3.14$

Fill Ratio = $\frac{\text{Cable Area}}{\text{Conduit Interior Area}} = \frac{\text{Cable } \pi * r^2}{\text{Conduit } \pi * r^2}$

Wrapping Tube Cable (WTC) Preparing the Cable End for Jetting and Blowing

Preparing the Cable End (Jetting)

Cable Carrier (A) and Parachute (B) (Jetting Methods)

Both the **cable carrier** (A) and **parachute** (B) methods are typically used in **larger duct applications**.

Typical Method for attaching the **Cable Carrier** or **Parachute** device:

- Apply the proper sized **fiber optic pulling grip** over the cable.
- Attach a **600 lbs. break-away** swivel to the end of the pulling grip.
- Attach the **Cable Carrier** or **Parachute** to the break-away swivel.
- Cover the **pulling grip** with electrical tape to help reduce the friction within the duct.



Preparing the Cable End (Blowing)

Cable Tips: (Blowing)

- When possible, **use a cable tip to seal the end** of the cable prior to installing process.
- Sealing the end will help eliminate the **air** or **lubricant** from entering the cable during the installation process.
- Cable **tips** will also help to **prevent** the **cables edge** from **catching** on the **couplers** within the duct and should help provide a smoother overall installation.
- **Tightly** place and **secure a cable tip** over the outer jacket so that it will **not** become **detached during the installation process**.



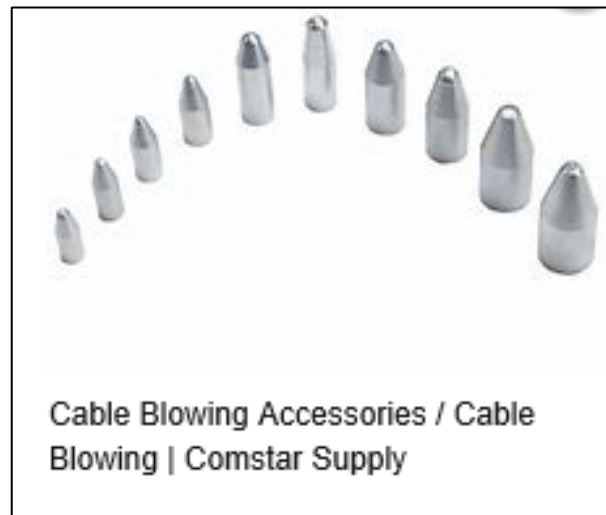
Preparing the Cable End (Equipment)



Carrier for **Jetted** Cable



Inflatable Carrier Parachute / **Jetted** Cable



Cable Tips for **Blown** Cable

Example of Carriers: (Jetted Cable)

<https://www.comstarsupply.com/catalogsearch/result/?q=carrier>

Example of Inflatable Carrier Parachute: (Jetted Cable)

<https://www.gmptools.com/line-carriers/>

<https://www.comstarsupply.com/catalogsearch/result/?q=carrier>

Example of Cable Tips: (Blown Cable)

<https://www.condux.com/product.php?id=426>

https://www.linemen-tools.com/category_s/723.htm

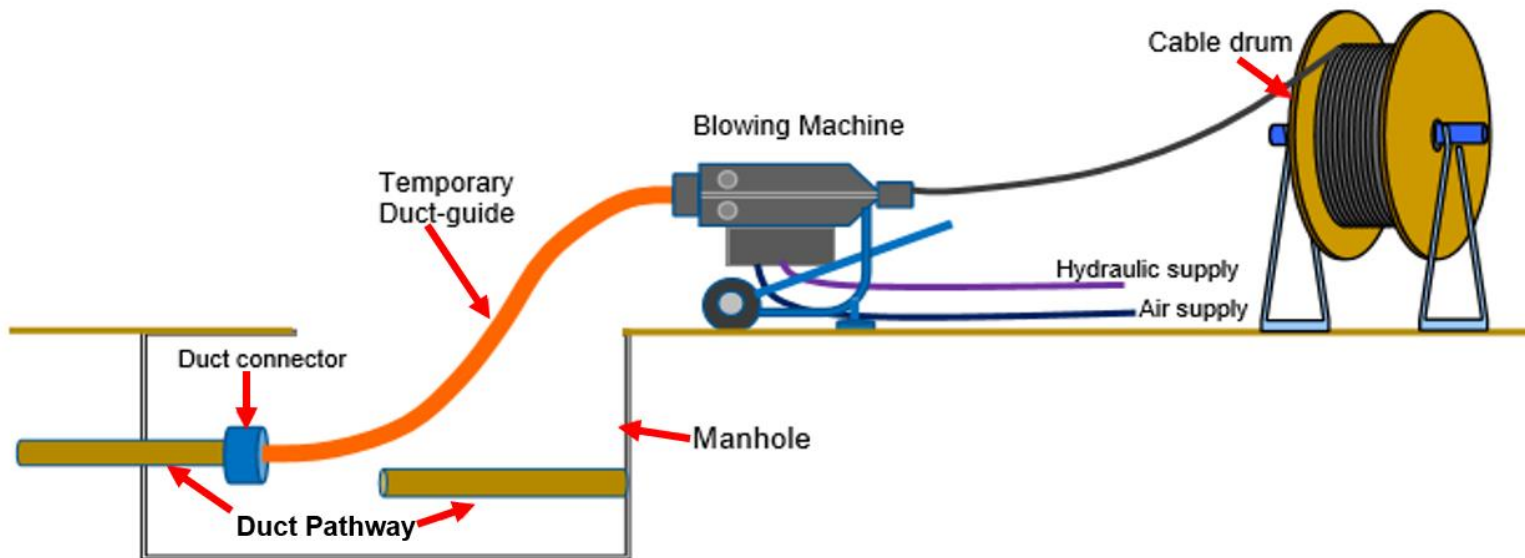
Last know vendor for large Cable Tips was Comstar Supply: <https://www.comstarsupply.com>

Wrapping Tube Cable (WTC) Air Assisted Installations Methods End to End, Mid-Point and Cascading

End to End Method (Air Assisted)

This method is used to install a continuous length of cable **between two given points**.

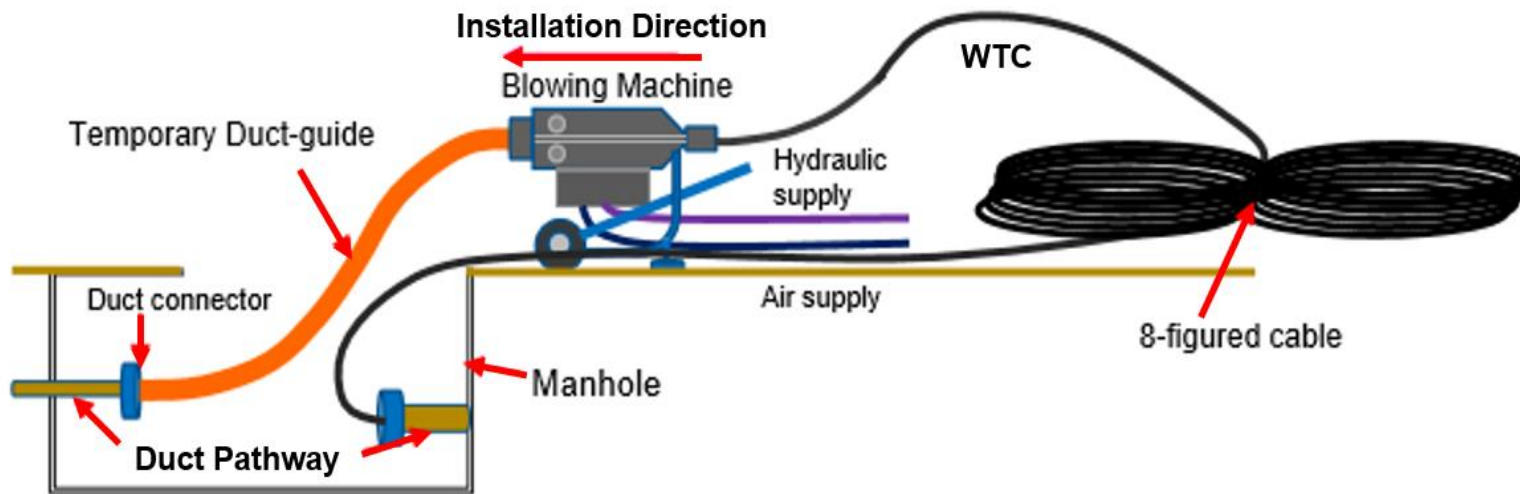
End to End Method - Air Assisted Installation:



Mid-Point / Figure 8 Method (Air Assisted)

Mid-point cable installation is used to **install a continuous length of cable**, which would otherwise **not be possible** from single end location.

Mid-Point / Figure 8 Method- Air Assisted Installation:

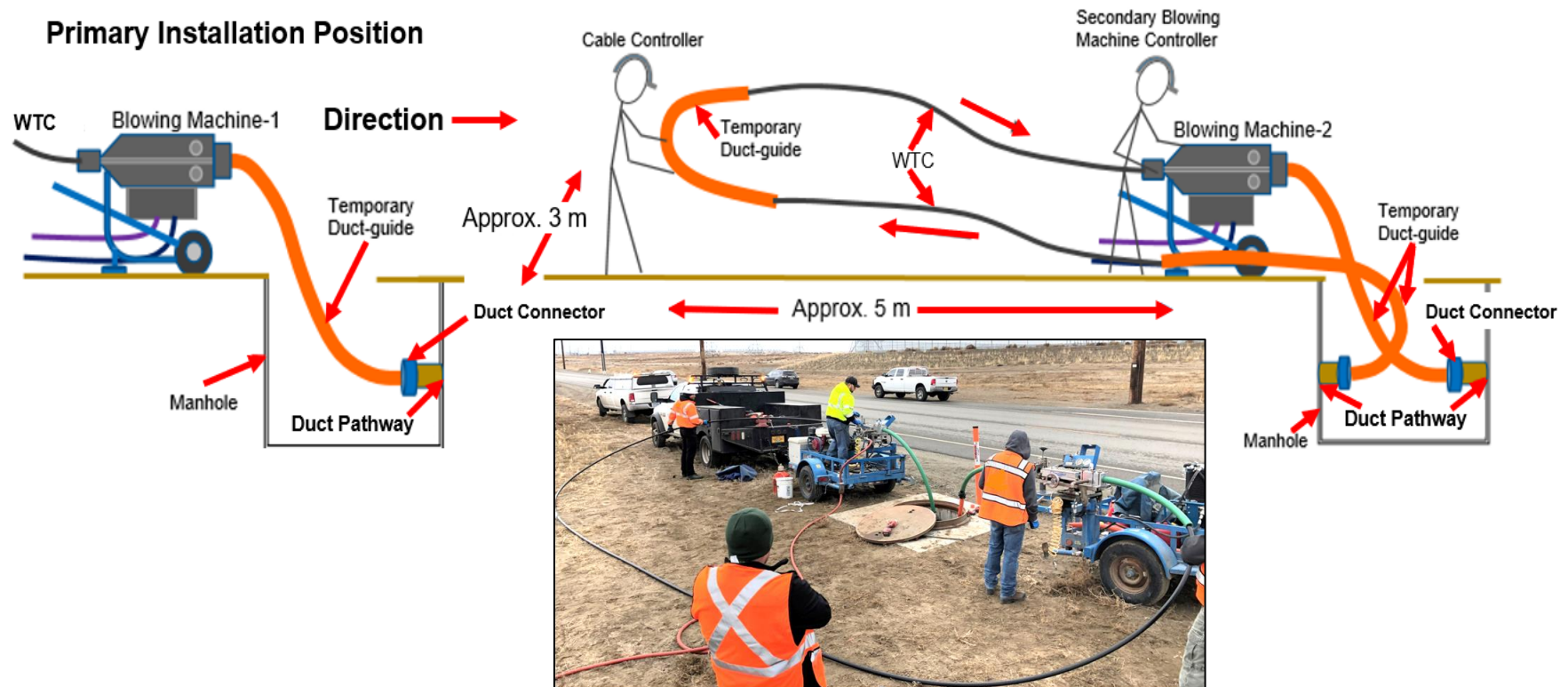


Note: The detailed methods for both the Figure 8 and Racetrack procedures / video are located on pages **80 – 89** of this document.

Cascade / Tandem Method (Air Assisted)

This method is used to install a continuous length of cable **twice the distance** that typically achieved by a **single point installation**.

Cascade / Tandem Method- Air Assisted Installation:



Wrapping Tube Cable (WTC)

Key Points for Post Air Assisted Installations

Key Points for Post Air Assisted Installations

Post Air Assisted Installations:

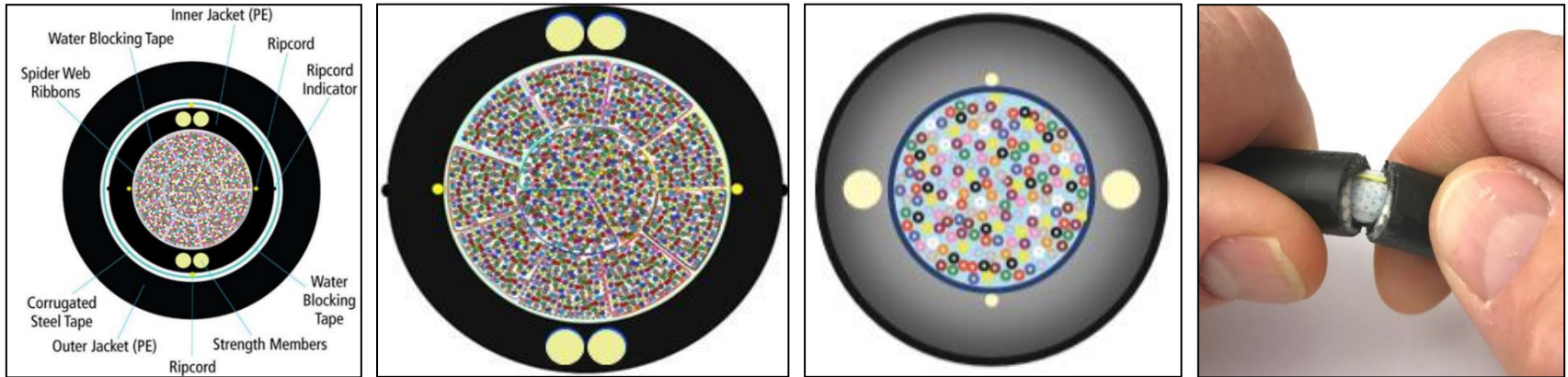
- **Leave enough cable** on each end to reach the intended application.
- Use one of the **AFL Recommended Methods to Coil the WTC**. **Never** use the **Fold and Flip** procedure to store the cable.
- **Secure all cable** against potential damage.
- **Identify and label** all cables for future applications as needed.
- **If required: Document the cable lengths, routing direction, fiber type and cable ID.**
- **Secure any open areas.**



AFL Coiling Method

Wrapping Tube Cable (WTC) Figure 8 and Racetrack Coiling Methods

Figure 8 Coiling Method



Natural or Preferential Bend of the WTC:

The WTC cable has **two** (1X1) or **four** (2 x 2) **longitudinal strength members** as shown in the cross-sectional diagram above. The strength member orientation creates a preferential bend property within the cable. Because of this bending feature, installation crews need to **know the proper methods for handling the cable prior to it being installed**. If the cable is **forced to bend against its preferential bend**, the torque within the cable may induce a twist into the outer jacket. If the outer jacket twist is excessive (**>180 degrees in one meter**), then the optical fibers within that section can potentially show changes in the optical performance.

Figure 8 Coiling Method

Figure-8 Coiling:

It is not uncommon during an underground construction project to **have to coil the cable off the reel in order to pass through an obstruction**. This is called **unreeling and back-feeding** the cable. A safe method of doing this is to “**Figure-8**” the cable off the reel. When properly performed, the figure-8 method will provide a means to unreel significant quantities of cable for passing through an obstruction. **The overall size of the Figure-8 coil is dependent on the cable diameter and the amount of cable to be back-feed**. Using a larger figure-8 is typically better for speed but will require more area to perform.

To **build a Figure-8 coil**, start with **two traffic cones spaced 30 feet apart** as the apex of each coil. Between the two end cones, **place two additional cones side by side for the cross over point** of the figure-8.



Figure 8 Coiling Method

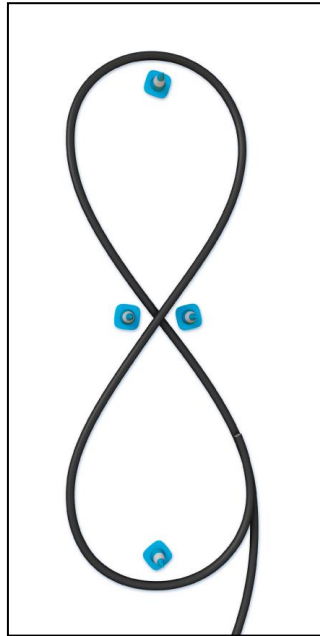
Figure-8 Coiling:

When forming a Figure 8, work **with the natural bend** of the WTC cable in order to **avoid inducing any twist** within the individual loops. After forming a loop, the cable should be routed through the two center cones and placed in the opposite direction of the loop prior. **(Clockwise and Counterclockwise)**

Keep the crossover point at a reasonable height that can be easily flipped for installation. If that height is reached, a **second figure-8** can be formed adjacent to the first.



Single Figure-8 “Example”



Double Figure-8 “Example”

Figure 8 Coiling Method

Allow the Cable to Lay Flat Naturally without Inducing Any Torque or Twist:

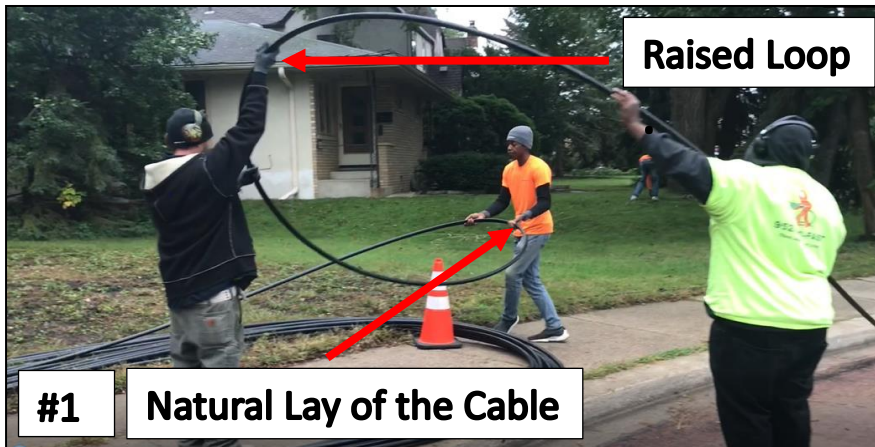
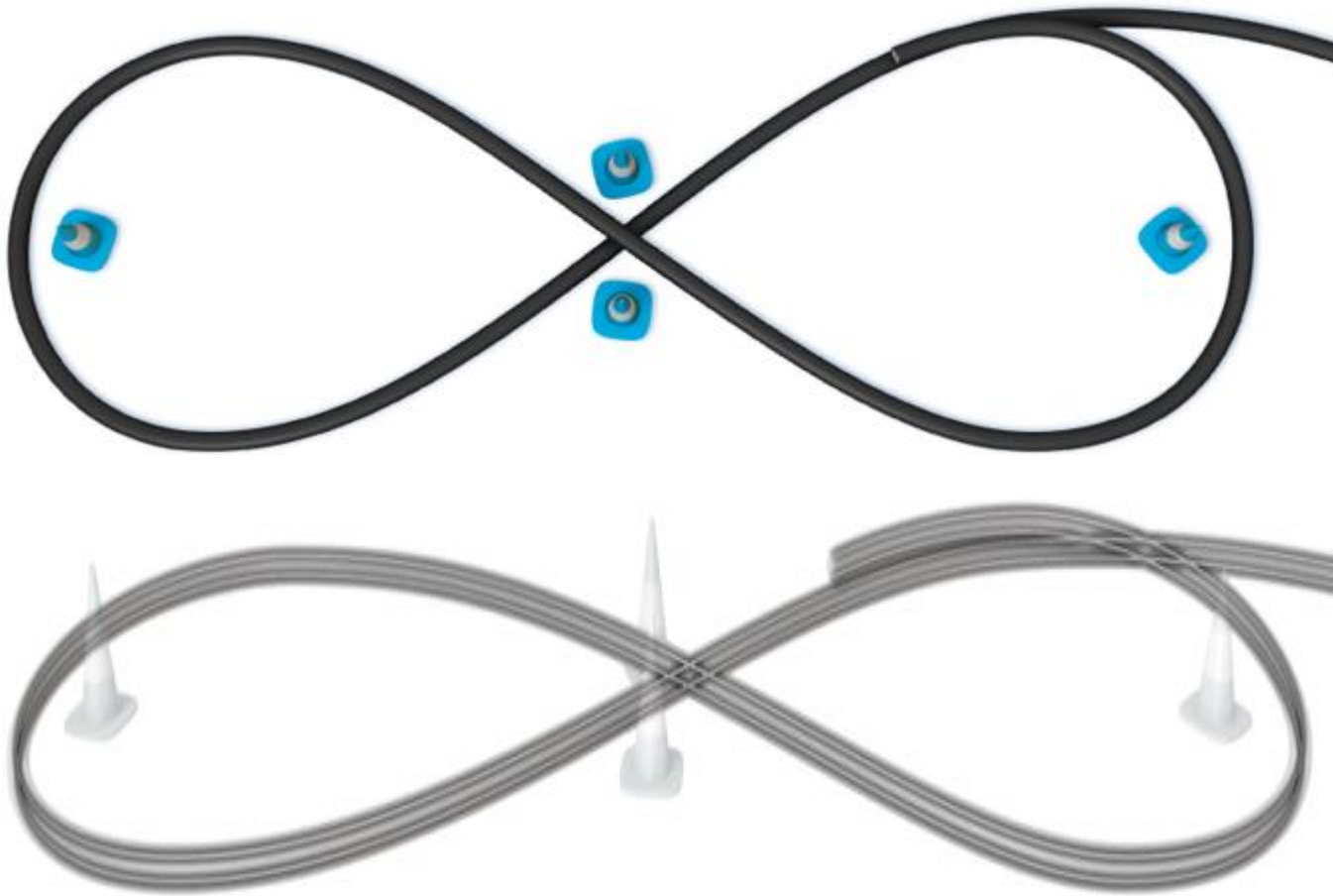


Figure 8 (Diagram)

Top and Side Angle (Showing the FRP Rod Positions)



Racetrack Coiling Method

Racetrack Coiling Method:

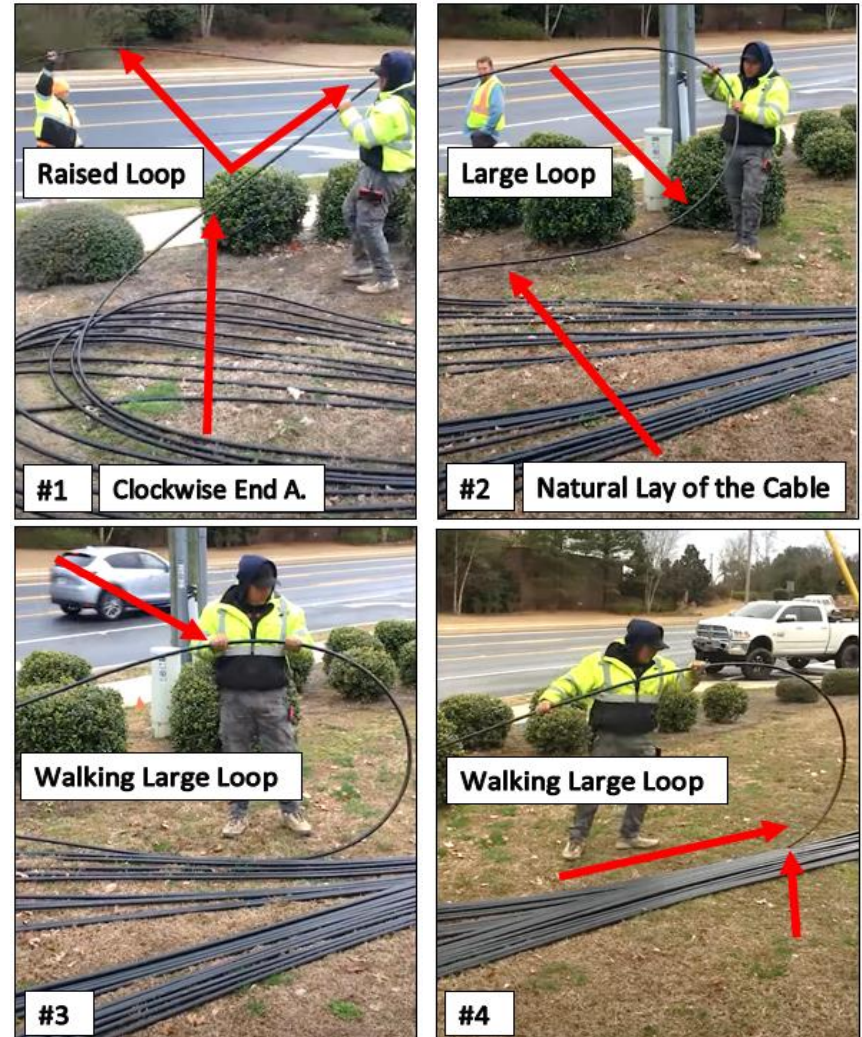
The racetrack method is an alternative to the figure-8 process. This process requires a **longer available footprint** to place the cable.

Mid-Point Method:

Route the cable as shown with each loop being layered and offset **on top of the loop** below it. This process will allow the cable to be fed back into the installation machine without flipping the cable.

End to End Method:

Route the cable in the same configuration as shown with each loop being layered and offset **under the loop** above it. This process will allow the cable to be fed back into the installation machine without flipping the cable.



Mid-Point Method Shown

Racetrack Coiling Method

Mid-Point Method Shown

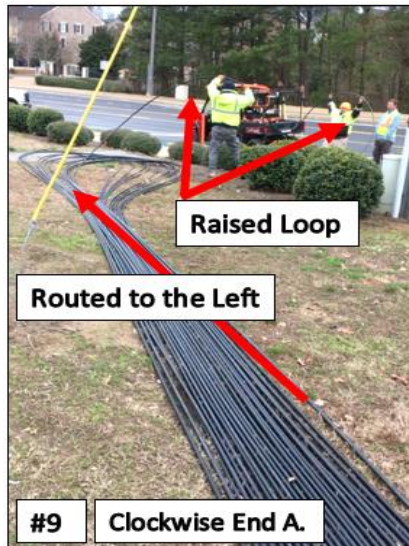
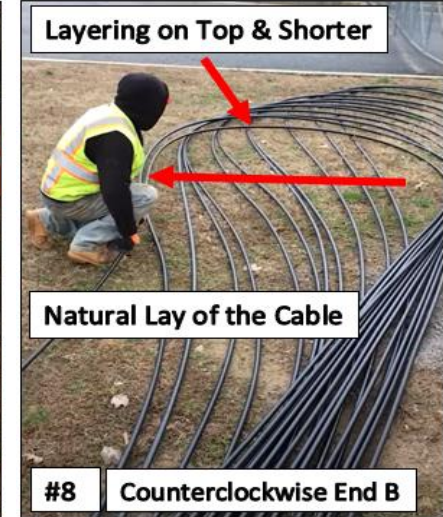
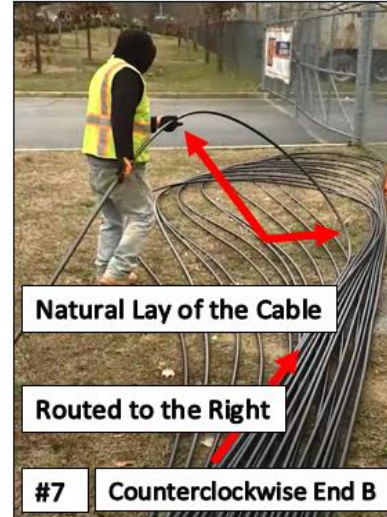


Figure 8 and Racetrack Coiling Methods



Figure 8 Example



Racetrack Example

Figure 8 – Racetrack (Video)

Wrapping Tube Cable (WTC) Coiling WTC for Storage

Coiling WTC for Storage

Coiling WTC for Storage: (Reference page 122 for the video link)

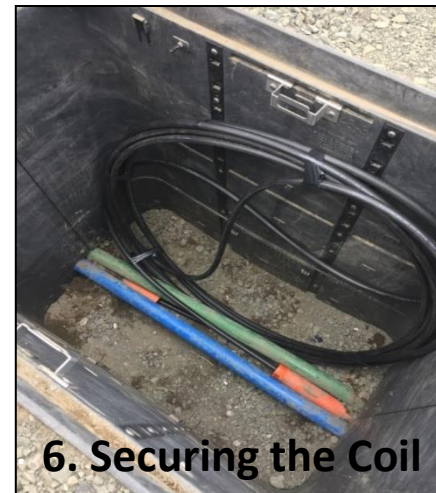
Prior to coiling the cable for storage, review the WTC specification sheet and understand the **Minimum Bend Radius** for **Long Term** storage and how that corresponds to each storage method. There are several **coiling** and **handling techniques** shown in the AFL video entitled “**Coiling Wrapping Tube Cable (WTC) for Storage**”. Each procedure shown, address different situations where one method of handling the cable may work better than another. Regardless of which method is used, care must be taken to avoid **inducing twist** into the jacket during the coiling process.

Standard AFL Coiling Method: (Reference page 90 – 96)

AFL recommends that you remove and extend the cable out from the storage area until you can identify the center of the loop. To form a coil, start by **aligning the strength members horizontally** on both the **upper and lower** portion of the loop. Using the AFL minimum bend radius specification, form and tape an **initial teardrop** loop at the cable's center point. Begin coiling the cable loop towards the storage area while maintaining the natural bend of the cable and minimum bend radius. If signs of twist begin to develop during this process, you can **stop and flip the entire coil** in the opposite direction to remove the twist. Once the coil has been lowered into the storage area, you can adjust its overall size while remaining within the specified diameter or larger. Once the coil is adjusted, it can be tie wrapped or taped together as needed.

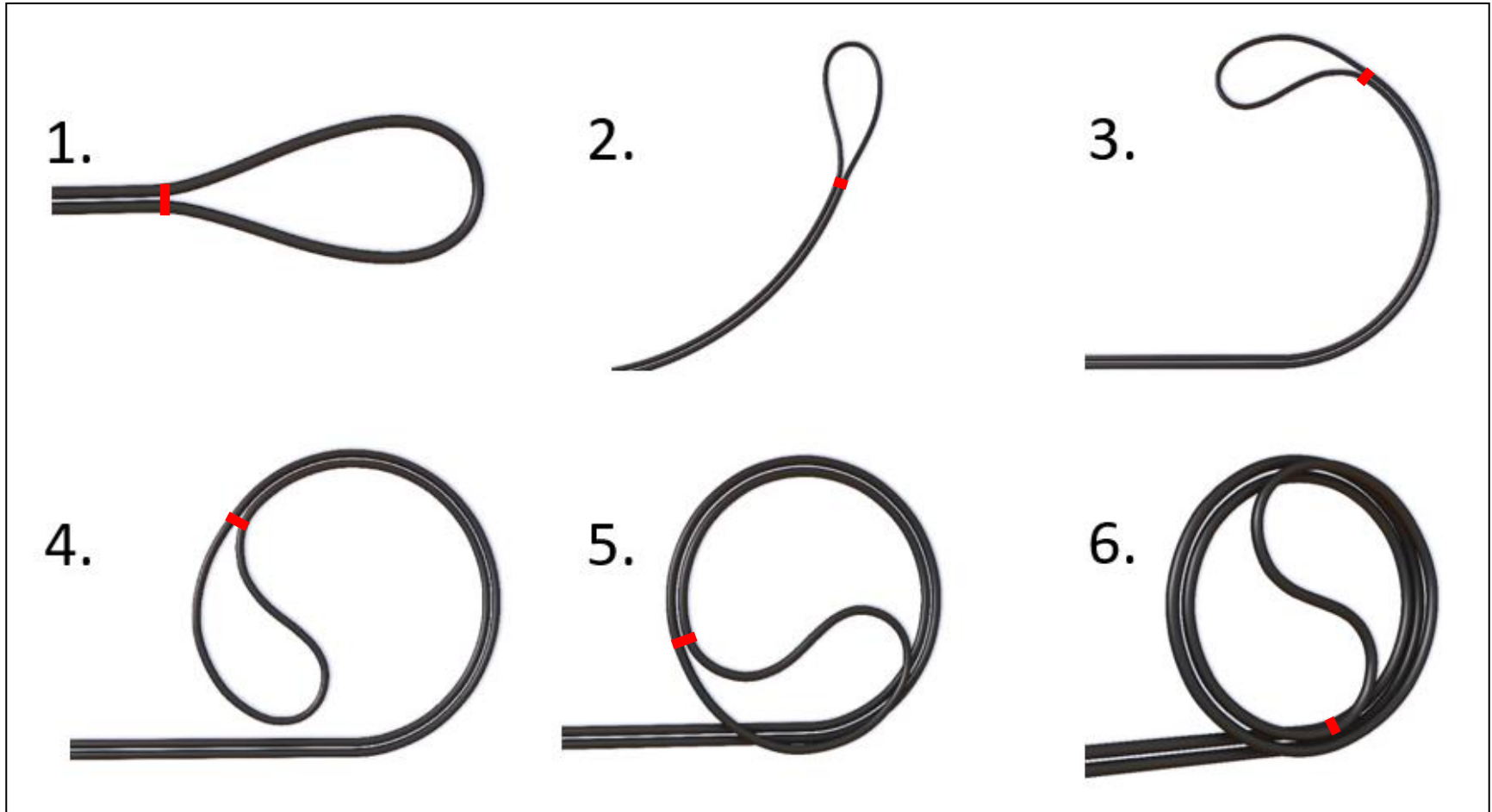
Coiling WTC for Storage

Standard AFL Coiling Method:



Coiling WTC for Storage

Standard AFL Coiling Method: (Coil Under or Over – Per Location)



Coiling WTC for Storage

Standard AFL Coiling Method: (Hand Hole / Vault Shown)



Incorrect Method – **Cross and Flip** (Twist)

Correct Method – **Roll Up** (No Twist)

Coiling WTC for Storage

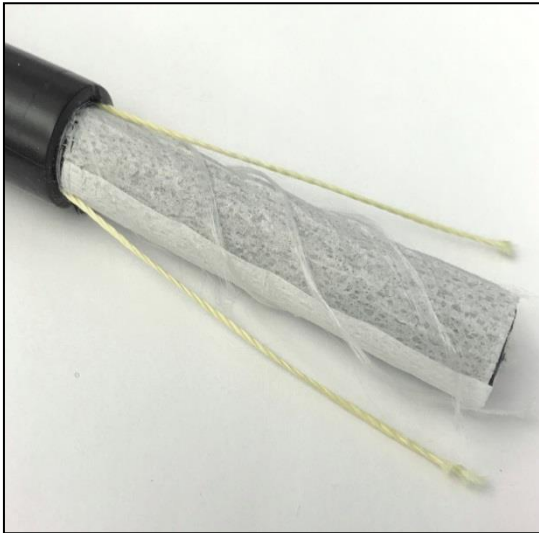
Coiling WTC Method: (Manhole)



Coiling WTC for Storage (Methods: Video)

Wrapping Tube Cable (WTC) Cable End Preparation

End Cable Preparation



Written Procedures are Available on the AFL Thumb Drives
(Reference page 120 for the video link)

End Cable Preparation Video



AFL WTC Tools: Cable End Preparation

The tools and materials that may be used to perform the WTC End Preparation procedure are shown in Figure 1. These materials are typically considered to be "Standard Tools" by field technicians for entering Armored and Non-Armored cables.



Safety Glasses
Safety Gloves
Cable Cutter
Razor Knife
Sharp Cable Knife
Black Marker

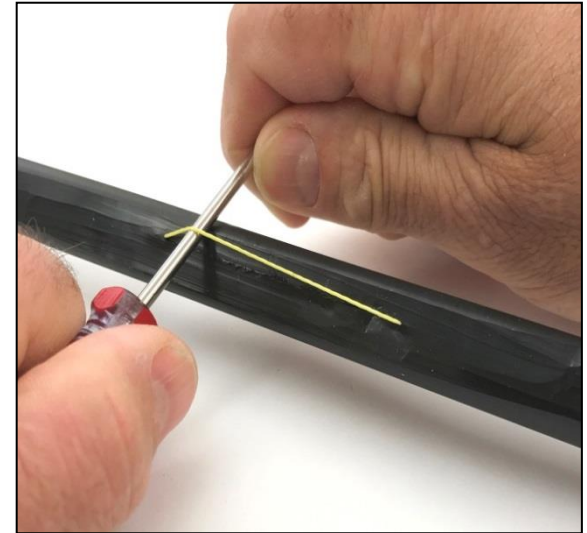
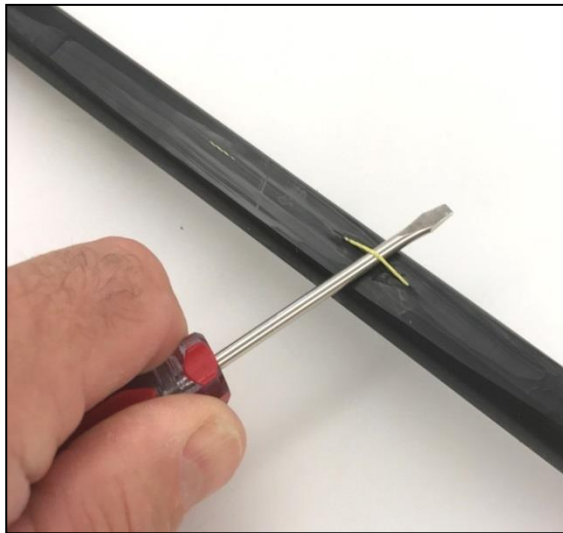
Paint Marker
Armor Entry Tool
Lineman Pliers
Large Screwdriver
Allen Wrench
Seam Ripper

Diagonal Cutting Pliers
Electrical or Vinyl Tape
Tape Measure
Splicing Scissors
Tubing Cutter

Figure 1. Tools and Materials for WTC Preparation.

Wrapping Tube Cable (WTC) Mid-Sheath Entry Preparation

Mid-Sheath Preparation



Written Procedures are Available on the AFL Thumb Drives
(Reference page 120 for the video link)

Mid-Sheath Preparation Video



Mid-Sheath Entry (Orientation)

WTC Cable and Fiber Binder Group Orientation

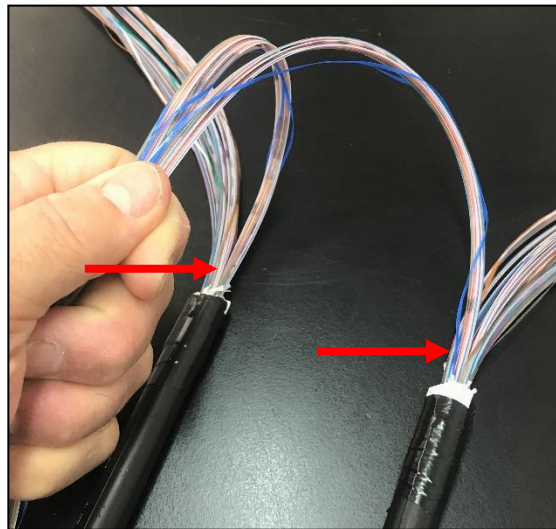
When placing the Mid-Sheath section of WTC into a splice closure, perform the following steps to assure the best results for accessing the Fiber Binder Groups.

1. Place the **cable ridge** in the **upward position** (1) to allow both the cable and closure to be easily rolled into the final storage location.
2. To verify that both ridges are in the correct position, place the top binder group next to the end of the cable (2). If the binder groups **don't reach** the same position (3), **rotate the cable** before securing it into the closure.

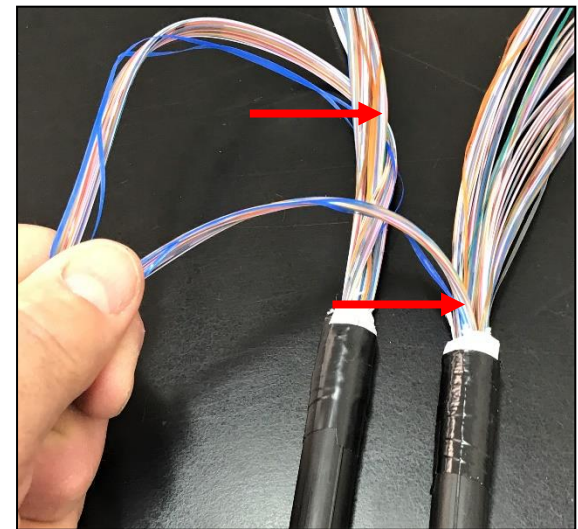
1. Ridge Up Orientation



2. Correct Binder Orientation



3. Incorrect Binder Orientation



AFL WTC Tools: Mid-Sheath Entry

The tools and materials that may be used to perform the WTC Mid-Sheath Entry procedure are shown in Figure 1. These materials are typically considered to be "Standard Tools" by field technicians for entering Armored and Non-Armored cables.



Safety Glasses

Safety Gloves

Cable Cutter

Razor Knife or

Sharp Cable Knife

Black Marker or

Paint Marker

Armor Entry Tool

Screwdriver - Large

Screwdriver - Small

Allen Wrench

Seam Ripper

Electrical and Vinyl Tape

Tape Measure

Splicing Scissors

Needle Nose Pliers

Flat Nose Pliers

Lineman Pliers

Diagonal Cutting Pliers

Tubing Cutter

Mid-Sheath Entry Tool

Figure 1. Tools and Materials for WTC Preparation.

Wrapping Tube Cable (WTC) Prepping Supplies

WTC Cable Entry Tool



Wrapping Tube Cable (WTC) Cable Entry Tool

This tool is designed to help efficiently open Wrapping Tube Cable in both end prep and mid sheath entry. The blade angle and tool shape help prevent the blade from cutting too deep.

Ordering Information

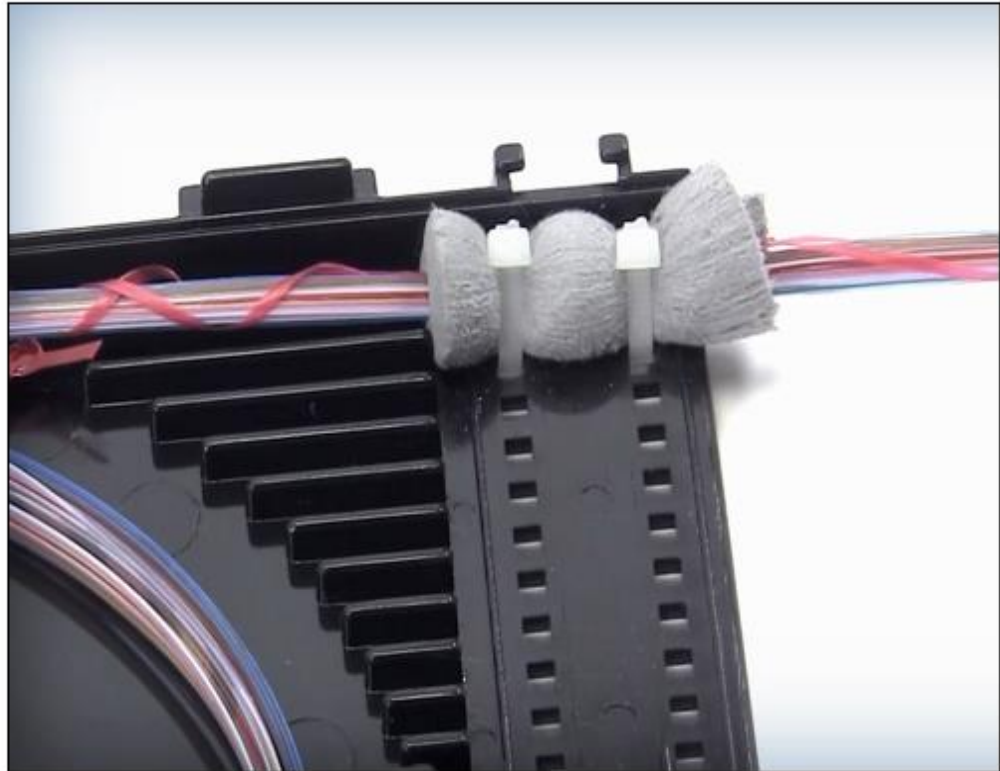
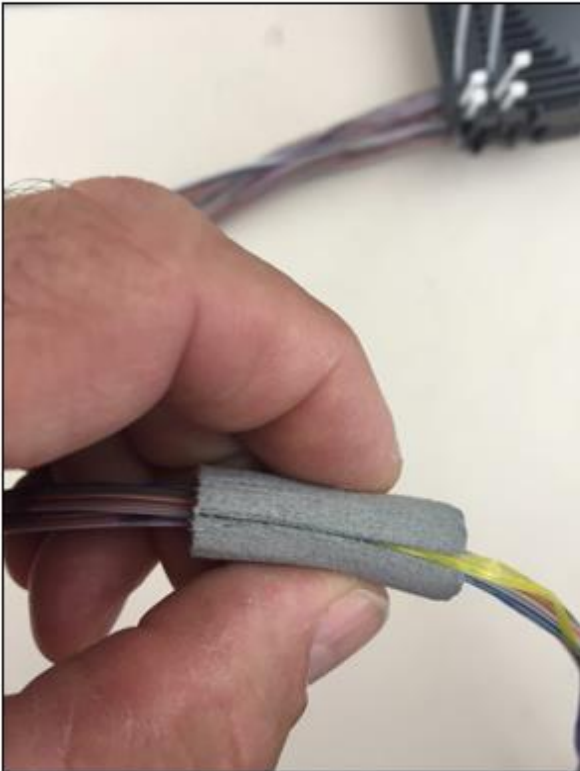
DESCRIPTION	AFL NO.
WTC Cable Entry Tool	CS014530
Replacement Blade for WTC Entry Tool	CS014966

WTC: Bundle Splice Tray Retention Kit

WTC-SWR Bundle Splice Tray Retention Kit:

Includes 25 foam grommets for retaining SWR bundles to splice trays

Foam Retention Kit: HW000406

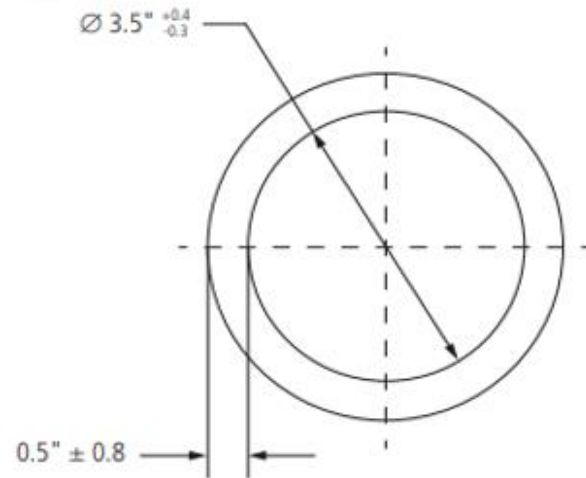


WTC: Silicone Spiral Wrap

Silicone Spiral Wrap

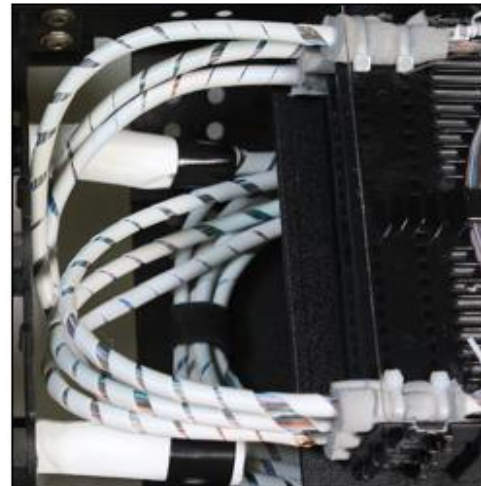
Silicone spiral wrap can be used to protect SpiderWeb Ribbon® bundles (up to 288 fibers) as well as up to 12 standard ribbons. The silicone spiral wrap can protect expressed fiber slack in various applications as well as act as a transition tube from a slack storage basket to splice trays in various splice closures.

Dimensions



Ordering Information

DESCRIPTION	AFL NO.
Silicone Spiral Wrap, 5.5 Foot Length	FC001657



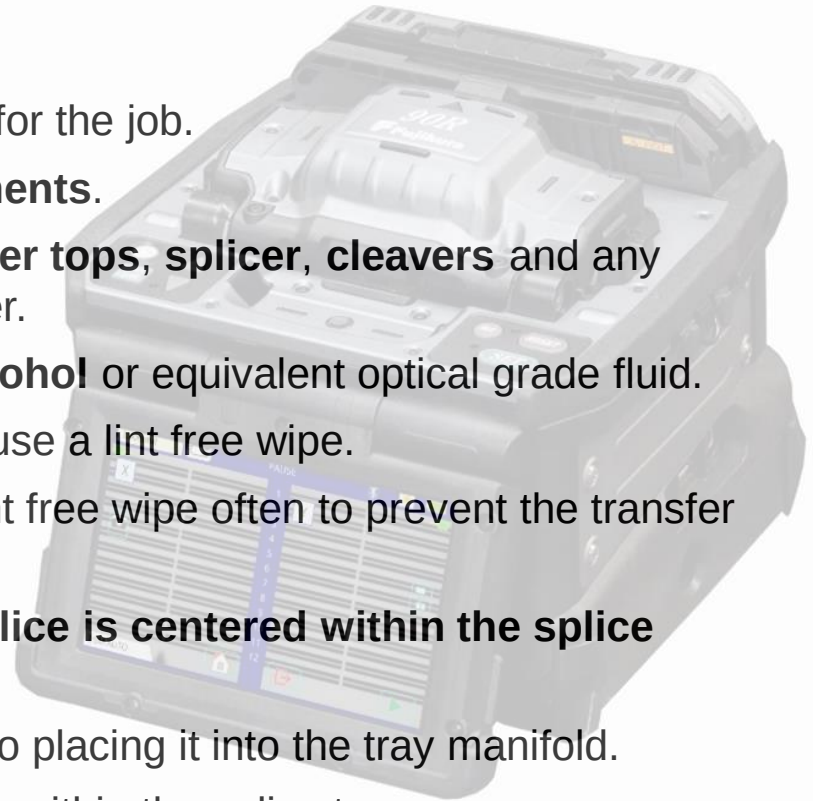
SWR Fiber Splicing and Equipment Guidelines

SWR Fiber Splicing Guidelines

General Splicing Guidelines:

AFL WTC Cable: 144 to 6912 fibers (SWR):

- Have the **proper fiber splicing** equipment for the job.
- Know the **customers splice loss requirements**.
- Always **maintain cleanliness** on the **counter tops, splicer, cleavers** and any other areas that may contact the optical fiber.
- When splicing, **use a 97%(+) isopropyl alcohol** or equivalent optical grade fluid.
- When **cleaning** the **optical fibers**, always use a lint free wipe.
- When splicing **ribbon fibers**, replace the lint free wipe often to prevent the transfer of debris.
- After splicing, make sure that the **fusion splice is centered within the splice sleeve prior to heating**.
- Allow the **splice sleeve time to cool** prior to placing it into the tray manifold.
- **Maintain the fibers minimum bend radius** within the splice tray.
- **Secure the sleeves** within the tray.
- **Check for bends or loose fibers** prior to **securing the lid** to the tray.



Splicing Equipment Guidelines

Setup Parameters of 70R+ or 90R:

The fusion splicer **must be set to splice 12 fibers using the correct SM splice setting.**

An **Arc Calibration** test should be performed regularly as needed. Other instances include, **after shipping, changes in temperature and elevation variations.**

The **heater** needs to be set for the **40 mm - 12 fiber mass splice protection sleeve.**

Setup of Thermal Strippers:

A high **heat setting** on the thermal stripper is best for stripping Spider Web Ribbon and others. Use a setting of **4** for the **AFL RS series.**



Splicing Equipment Guidelines

Cleaning the Fusion Splicer: (1)

The splicer and its **v-grooves** must be **free of dust and debris**.

Cleaning the Fiber Holders: (2)

The fiber holder and **retention pads** must be **clean** and free of **dust and debris**.

Cleaning the Thermal Stripper: (3)

The thermal stripper needs to be thoroughly **cleaned before starting** a project as well as **cleaned after each use**. Ensure there is **no dust or debris** on the **fiber holder channel**, the **blades**, and especially the **stripping channel** and **retention pads**.



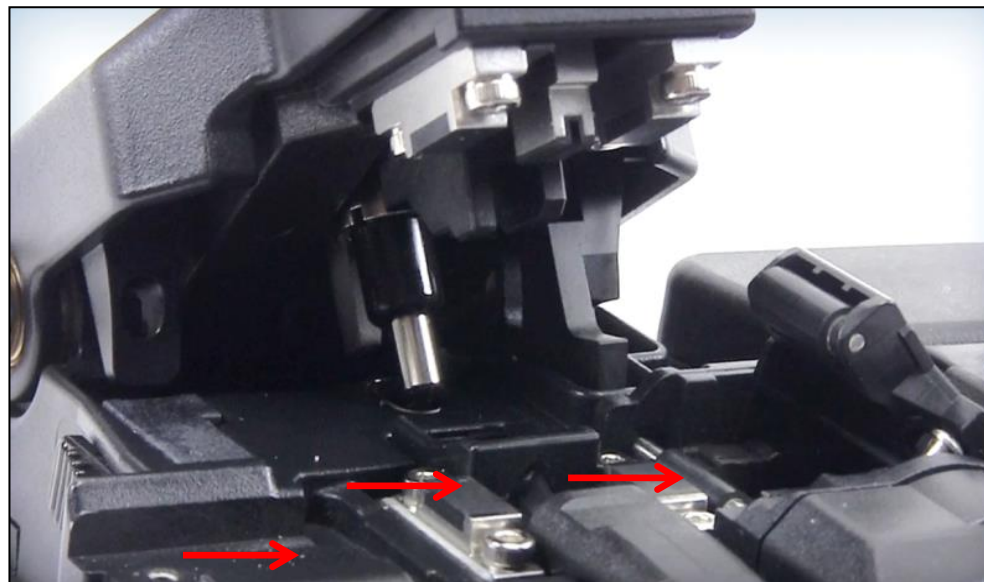
Splicing Equipment Guidelines

Cleaning the Cleaver:

Remove all **dust** and **debris** from the **fiber cleaver**. Clean the fiber **holder channel**, the top and bottom **retention pads**, and the **fiber rollers**.

Setting up the Cleaver:

Be sure the cleaver is **activated** and **ready to cleave**. Inspect the cleaver **blade** region to ensure it is **free of fiber shards**.



Cleaver Blade Rotation: (CT-30)

Accurate counts of cleaves should be noted before rotating or replacing the cleaver blades. (**83 x 12F** per rotation)

Cleaver Blade Rotation: (CT-50)

Rotated **manually** or by **Bluetooth**.



SWR Fiber Splicing Video (Mass Fusion)



AFLglobal.com

AFL
1 (800) 235-3423

Splicer Hotline
1 (800) 866-3602

Fiber Ribbonizing Tool (AFL RT-02)

AFL RT-02 Fiber Ribbonizing Tool:



Features

- No glue required
- 200 μ m and 250 μ m compatible
- Loading with color code sequence not required
- Fibers load directly into fiber holder
- Left and right fiber holder color codes printed on tool

Applications

- Ribbonizing 200 μ m and 250 μ m loose fibers
- 200 μ m and 250 μ m MPO termination
- Mass fusion splicing loose fiber cables

Ordering Information

DESCRIPTION	AFL NO.
RT-02 (tool only)	S017465
FH-70-12PC (pair of pitch conversion holders for 200 μ m loose fibers)	S017464
FH-70-12 (pair – standard 12F ribbon holders)	S017119

Splice Closure Recommendations

Splice Closure Recommendations

WTC Closure Recommendations:

- Make sure that the **splice closure** is the **correct one for the application**.
- **Before entering the cable**, be sure that the **splice closure has** the needed accessories: **splice trays, tie wraps, heat shrinks** etc.
- When placing the Wrapping Tube Cables (WTC) into the closure entry ports, be sure to secure each cable in the same specified orientation. (Ref. page 104)
- Make sure that **all fiber units** placed in the splice closure **meet** the specified **minimum bend radius requirements**.
- Make sure that the **closure is free of debris** prior to it being sealed.
- **Pressure testing** the closure if required. / **Release the pressure**.
- **Ground the closure** if required.
- **Follow the AFL coiling and cable storage** recommendations.



Questions and Answers

AFL Reference Material

Reference Material: (YouTube Videos)

AFL WTC Product Information, Video Links and Related Documents:

[Wrapping Tube Cable \(aflglobal.com\)](#)

Fiber Identification Charts for 288 – 864 WTC: (Page 14)

[Wrapping Tube Cable WTC with SWR \(aflglobal.com\)](#)

Figure 8 Storage of the AFL Wrapping Tube Fiber Optic Cable: (Page 89)

<https://www.youtube.com/watch?v=IzTtVKeafa8&feature=youtu.be>

Coiling Wrapping Tube Cable (WTC) for Storage: (Page 96)

<https://www.youtube.com/watch?v=IrMrWR4ru70&feature=youtu.be>

Recommended End Cable Entry Procedures for AFL Wrapping Tube Cable (WTC): (Page 99)

<https://www.youtube.com/watch?v=zYZN81wZQ1I&feature=youtu.be>

Recommended Mid-Sheath Entry Procedures for AFL Wrapping Tube Cable (WTC): (Page 103)

https://www.youtube.com/watch?v=Z_FZHL7tR64&feature=youtu.be

Wrapping Tube Cable (WTC) Cable Entry Tool: (Page 107)

[Wrapping Tube Cable WTC Cable Entry Tool \(aflglobal.com\)](#)

Splicing AFL's Wrapping Tube Cable (WTC) with SpiderWeb Ribbon (SWR): (Page 115)

https://www.youtube.com/watch?v=trYr_Gfr3J4&t=39s

RT-02 Ribbonizing Tool: (Page 116)

[RT 02 Ribbonizing Tool \(aflglobal.com\)](#)

Thank You!



www.AFLglobal.com